MR750 Dual Drive Upgrade

OPERATING DOCUMENTATION

5670088
Revision 2.0


WARNING
STRONG MAGNETIC FIELD
NO PACEMAKERS*
NO NEUROSTIMULATORS*
NO CONDUCTIVE/METALLIC IMPLANTS*
Persons with pacemakers, neurostimulators or metallic implants must not enter this area.
Serious injury may result.
* In general, patients with conductive (e.g. metallic) implants are contraindicated for MR scans. For patients with implants that are labeled as 'MR Safe' or 'MR Conditional', consult the implant device manufacturer's documentation.
* WARNING: Only use quadrature transmit for 'MR Conditional' devices.
5457590

Important Information

LANGUAGE

ПРЕДУПРЕЖДЕНИЕ (BG)	Това упътване за работа е налично само на английски език. • Ако доставчикът на услугата на клиента изиска друг език, задължение на клиента е да осигури превод. • Не използвайте оборудването, преди да сте се консултирали и разбрали упътването за работа. • Неспазването на това предупреждение може да доведе до нараняване на доставчика на услугата, оператора или пациента в резултат на токов удар, механична или друга опасност.
警告 (ZH-CN)	本维修手册仅提供英文版本。 • 如果客户的维修服务人员需要非英文版本，则客户需自行提供翻译服务。 • 未详细阅读和完全理解本维修手册之前，不得进行维修。 • 忽略本警告可能对维修服务人员、操作人员或患者造成电击、机械伤害或其他形式的伤 害。
警告 (ZH-HK)	本服務手冊僅提供英文版本。 • 倘若客戶的服務供應商需要英文以外之服務手冊，客戶有責任提供翻譯服務。 • 除非已參閱本服務手冊及明白其內容，否則切勿嘗試維修設備。 • 不遵從本警告或會令服務供應商、網絡供應商或病人受到觸電、機械性或其他的危險。
警告 (ZH-TW)	本維修手冊僅有英文版。 • 若客戶的維修廠商需要英文版以外的語言，應由客戶自行提供翻譯服務。 • 請勿試圖維修本設備，除非 您已查閱並瞭解本維修手冊。 • 若未留意本警告，可能導致維修廠商、操作員或病患因觸電、機械或其他危險而受傷。
UPOZORENJE (HR)	Ovaj servisni priručnik dostupan je na engleskom jeziku. • Ako davatelj usluge klijenta treba neki drugi jezik, klijent je dužan osigurati prijevod. • Ne pokušavajte servisirati opremu ako niste u potpunosti pročitali i razumjeli ovaj servisni priručnik. • Zanemarite li ovo upozorenje, može doći do ozljede davatelja usluge, operatera ili pacijenta uslijed strujnog udara, mehaničkih ili drugih rizika.

VÝSTRAHA (CS)

Tento provozní návod existuje pouze v anglickém jazyce.

- V případě, že externí služba zákazníkům potřebuje návod v jiném jazyce, je zajištění překladu do odpovídajícího jazyka úkolem zákazníka.
- Nesnažte se o údržbu tohoto zařízení, aniž byste si přečetli tento provozní návod a pochopili jeho obsah.
- V případě nedodržování této výstrahy může dojít k poranění pracovníka prodejního servisu, obslužného personálu nebo pacientů vlivem elektrického proudu, respektive vlivem mechanických či jiných rizik.

ADVARSEL (DA)

Denne servicemanual findes kun på engelsk.

- Hvis en kundes tekniker har brug for et andet sprog end engelsk, er det kundens ansvar at sørge for oversættelse.
- Forsøg ikke at servicere udstyret uden at læse og forstå denne servicemanual.
- Manglende overholdelse af denne advarsel kan medføre skade på grund af elektrisk stød, mekanisk eller anden fare for teknikeren, operatøren eller patienten.

WAARSCHUWING (NL)

Deze onderhoudshandleiding is enkel in het Engels verkrijgbaar.

- Als het onderhoudspersoneel een andere taal vereist, dan is de klant verantwoordelijk voor de vertaling ervan.
- Probeer de apparatuur niet te onderhouden alvorens deze onderhoudshandleiding werd geraadpleegd en begrepen is.
- Indien deze waarschuwing niet wordt opgevolgd, zou het onderhoudspersoneel, de operator of een patiënt gewond kunnen raken als gevolg van een elektrische schok, mechanische of andere gevaren.

WARNING (EN)

This service manual is available in English only.

- If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services.
- Do not attempt to service the equipment unless this service manual has been consulted and is understood.
- Failure to heed this warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.

HOIATUS (ET)

See teenindusjuhend on saadaval ainult inglise keeles.

- Kui klienditeeninduse osutaja nõuab juhendit inglise keelest erinevas keeles, vastutab klient tõlketeenuse osutamise eest.
- Ärge üritage seadmeid teenindada enne eelnevalt käesoleva teenindusjuhendiga tutvumist ja sellest aru saamist.
- Käesoleva hoiatuse eiramine võib põhjustada teenuseosutaja, operaatori või patsiendi vigastamist elektrilöögi, mehaanilise või muu ohu tagajärjel.

VAROITUS (FI)

Tämä huolto-ohje on saatavilla vain englanniksi.

- Jos asiakkaan huoltohenkilöstö vaatii muuta kuin englanninkielistä materiaalia, tarvittavan käännöksen hankkiminen on asiakkaan vastuulla.
- Älä yritä korjata laitteistoa ennen kuin olet varmasti lukenut ja ymmärtänyt tämän huolto-ohjeen.
- Mikäli tätä varoitusta ei noudateta, seurauksena voi olla huoltohenkilöstön, laitteiston käyttäjän tai potilaan vahingoittuminen sähköiskun, mekaanisen vian tai muun vaaratilanteen vuoksi.

ATTENTION (FR)

Ce manuel d’installation et de maintenance est disponible uniquement en anglais.

- Si le technicien d'un client a besoin de ce manuel dans une langue autre que l'anglais, il incombe au client de le faire traduire.
- Ne pas tenter d'intervenir sur les équipements tant que ce manuel d’installation et de maintenance n'a pas été consulté et compris.
- Le non-respect de cet avertissement peut entraîner chez le technicien, l'opérateur ou le patient des blessures dues à des dangers électriques, mécaniques ou autres.

WARNUNG (DE)

Diese Serviceanleitung existiert nur in englischer Sprache.

- Falls ein fremder Kundendienst eine andere Sprache benötigt, ist es Aufgabe des Kunden für eine entsprechende Übersetzung zu sorgen.
- Versuchen Sie nicht diese Anlage zu warten, ohne diese Serviceanleitung gelesen und verstanden zu haben.
- Wird diese Warnung nicht beachtet, so kann es zu Verletzungen des Kundendiensttechnikers, des Bedieners oder des Patienten durch Stromschläge, mechanische oder sonstige Gefahren kommen.

ΠΡΟΕΙΔΟΠΟΙΗΣΗ (EL)

Το παρόν εγχειρίδιο σέρβις διατίθεται μόνο στα αγγλικά.

- Εάν ο τεχνικός σέρβις ενός πελάτη απαιτεί το παρόν εγχειρίδιο σε γλώσσα εκτός των αγγλικών, αποτελεί ευθύνη του πελάτη να παρέχει τις υπηρεσίες μετάφρασης.
- Μην επιχειρήσετε την εκτέλεση εργασιών σέρβις στον εξοπλισμό αν δεν έχετε συμβουλευτεί και κατανοήσει το παρόν εγχειρίδιο σέρβις.
- Αν δεν προσέξετε την προειδοποίηση αυτή, ενδέχεται να προκληθεί τραυματισμός στον τεχνικό σέρβις, στο χειριστή ή στον ασθενή από ηλεκτροπληξία, μηχανικούς ή άλλους κινδύνους.

FIGYELMEZTETÉS (HU)

Ezen karbantartási kézikönyv kizárólag angol nyelven érhető el.

- Ha a vevő szolgáltatója angoltól eltérő nyelvre tart igényt, akkor a vevő felelőssége a fordítás elkészíttetése.
- Ne próbálja elkezdeni használni a berendezést, amíg a karbantartási kézikönyvben leírtakat nem értelmezték.
- Ezen figyelmeztetés figyelmen kívül hagyása a szolgáltató, működtető vagy a beteg áramütés, mechanikai vagy egyéb veszélyhelyzet miatti sérülését eredményezheti.

AÐVÖRUN
(IS)

Þessi þjónustuhandbók er aðeins fáanleg á ensku.

- Ef að þjónustuveitandi viðskiptamanns þarfnast annas tungumáls en ensku, er það skylda viðskiptamanns að skaffa tungumálaþjónustu.
- Reynið ekki að afgreiða tækið nema að þessi þjónustuhandbók hefur verið skoðuð og skilin.
- Brot á sinna þessari aðvörun getur leitt til meiðsla á þjónustuveitanda, stjórnanda eða sjúklings frá raflosti, vélrænu eða öðrum áhættum.

AVVERTENZA
(IT)

Il presente manuale di manutenzione è disponibile soltanto in lingua inglese.

- Se un addetto alla manutenzione richiede il manuale in una lingua diversa, il cliente è tenuto a provvedere direttamente alla traduzione.
- Procedere alla manutenzione dell'apparecchiatura solo dopo aver consultato il presente manuale ed averne compreso il contenuto.
- Il mancato rispetto della presente avvertenza potrebbe causare lesioni all'addetto alla manutenzione, all'operatore o ai pazienti provocate da scosse elettriche, urti meccanici o altri rischi.

警告
(JA)

このサービスマニュアルには英語版しかありません。

- サービスを担当される業者が英語以外の言語を要求される場合、翻訳作業はその業者の責任で行うものとさせていただきます。
- このサービスマニュアルを熟読し理解せずに、装置のサービスを行わないでください。
- この警告に従わない場合、サービスを担当される方、操作員あるいは患者 さんが、感電や機械的又はその他の危険により負傷する可能性があります。

경고
(KO)

본 서비스 매뉴얼은 영어로만 이용하실 수 있습니다.

- 고객의 서비스 제공자가 영어 이외의 언어를 요구할 경우, 번역 서비스를 제공하는 것은 고객의 책임입니다.
- 본 서비스 매뉴얼을 참조하여 숙지하지 않은 이상 해당 장비를 수리하려고 시도하지 마십시오.
- 본 경고 사항에 유의하지 않으면 전기 쇼크, 기계적 위험, 또는 기타 위험으로 인해 서비스 제공자, 사용자 또는 환자에게 부상을 입힐 수 있습니다.

BRĪDINĀJUMS
(LV)

Šī apkopes rokasgrāmata ir pieejama tikai angļu valodā.

- Ja klienta apkopes sniedzējam nepieciešama informācija citā valodā, klienta pienākums ir nodrošināt tulkojumu.
- Neveiciet aprīkojuma apkopi bez apkopes rokasgrāmatas izlasīšanas un saprašanas.
- Šī brīdinājuma neievērošanas rezultātā var rasties elektriskās strāvas trieciena, mehānisku vai citu faktoru izraisītu traumu risks apkopes sniedzējam, operatoram vai pacientam.

ĮSPĖJIMAS (LT)	Šis eksploatavimo vadovas yra tik anglų kalba. • Jei kliento paslaugų tiekėjas reikalauja vadovo kita kalba – ne anglų, suteikti vertimo paslaugas privalo klientas. • Nemėginkite atlikti įrangos techninės priežiūros, jei neperskaitėte ar nesupratote šio eksploatavimo vadovo. • Jei nepaisysite šio įspėjimo, galimi paslaugų tiekėjo, operatoriaus ar paciento sužalojimai dėl elektros šoko, mechaninių ar kitų pavojų.
ADVARSEL (NO)	Denne servicehåndboken finnes bare på engelsk. • Hvis kundens serviceleverandør har bruk for et annet språk, er det kundens ansvar å sørge for oversettelse. • Ikke forsøk å reparere utstyret uten at denne servicehåndboken er lest og forstått. • Manglende hensyn til denne advarselen kan føre til at serviceleverandøren, operatøren eller pasienten skades på grunn av elektrisk støt, mekaniske eller andre farer.
OSTRZEŻENIE (PL)	Niniejszy podręcznik serwisowy dostępny jest jedynie w języku angielskim. • Jeśli serwisant klienta wymaga języka innego niż angielski, zapewnienie usługi tłumaczenia jest obowiązkiem klienta. • Nie próbować serwisować urządzenia bez zapoznania się z niniejszym podręcznikiem serwisowym i zrozumienia go. • Niezastosowanie się do tego ostrzeżenia może doprowadzić do obrażeń serwisanta, operatora lub pacjenta w wyniku porażenia prądem elektrycznym, zagrożenia mechanicznego bądź innego.
ATENÇÃO (PT-BR)	Este manual de assistência técnica encontra-se disponível unicamente em inglês. • Se outro serviço de assistência técnica solicitar a tradução deste manual, caberá ao cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • A não observância deste aviso pode ocasionar ferimentos no técnico, operador ou paciente decorrentes de choques elétricos, mecânicos ou outros.
ATENÇÃO (PT-PT)	Este manual de assistência técnica só se encontra disponível em inglês. • Se qualquer outro serviço de assistência técnica solicitar este manual noutro idioma, é da responsabilidade do cliente fornecer os serviços de tradução. • Não tente reparar o equipamento sem ter consultado e compreendido este manual de assistência técnica. • O não cumprimento deste aviso pode colocar em perigo a segurança do técnico, do operador ou do paciente devido a choques eléctricos, mecânicos ou outros.

ATENȚIE (RO)	Acest manual de service este disponibil doar în limba engleză. • Dacă un furnizor de servicii pentru clienți necesită o altă limbă decât cea engleză, este de datoria clientului să furnizeze o traducere. • Nu încercați să reparați echipamentul decât ulterior consultării și înțelegerii acestui manual de service. • Ignorarea acestui avertisment ar putea duce la rănirea depanatorului, operatorului sau pacientului în urma pericolelor de electrocutare, mecanice sau de altă natură.
ОСТОРОЖНО! (RU)	Данное руководство по техническому обслуживанию представлено только на английском языке. • Если сервисному персоналу клиента необходимо руководство не на английском, а на каком-то другом языке, клиенту следует самостоятельно обеспечить перевод. • Перед техническим обслуживанием оборудования обязательно обратитесь к данному руководству и поймите изложенные в нем сведения. • Несоблюдение требований данного предупреждения может привести к тому, что специалист по техобслуживанию, оператор или пациент получит удар электрическим током, механическую травму или другое повреждение.
UPOZORENJE (SR)	Ovo servisno uputstvo je dostupno samo na engleskom jeziku. • Ako klijentov serviser zahteva neki drugi jezik, klijent je dužan da obezbedi prevodi-lačke usluge. • Ne pokušavajte da opravite uređaj ako niste pročitali i razumeli ovo servisno uputst-vo. • Zanemarivanje ovog upozorenja može dovesti do povređivanja servisera, rukovaoca ili pacijenta usled strujnog udara ili mehaničkih i drugih opasnosti.
UPOZORNENIE (SK)	Tento návod na obsluhu je k dispozícii len v angličtine. • Ak zákazníkov poskytovateľ služieb vyžaduje iný jazyk ako angličtinu, poskytnutie prekladateľských služieb je zodpovednosťou zákazníka. • Nepokúšajte sa o obsluhu zariadenia, kým si neprečítate návod na obluhu a nepor-ozumiete mu. • Zanedbanie tohto upozornenia môže spôsobiť zranenie poskytovateľa služieb, ob-sluhujúcej osoby alebo pacienta elektrickým prúdom, mechanické alebo iné ohroze-nie.
ATENCION (ES)	Este manual de servicio sólo existe en inglés. • Si el encargado de mantenimiento de un cliente necesita un idioma que no sea el inglés, el cliente deberá encargarse de la traducción del manual. • No se deberá dar servicio técnico al equipo, sin haber consultado y comprendido este manual de servicio. • La no observancia del presente aviso puede dar lugar a que el proveedor de servi-cios, el operador o el paciente sufran lesiones provocadas por causas eléctricas, mecánicas o de otra naturaleza.

VARNING (SV)

Den här servicehandboken finns bara tillgänglig på engelska.

- Om en kunds servicetekniker har behov av ett annat språk än engelska, ansvarar kunden för att tillhandahålla översättningstjänster.
- Försök inte utföra service på utrustningen om du inte har läst och förstår den här servicehandboken.
- Om du inte tar hänsyn till den här varningen kan det resultera i skador på servicete-knikern, operatören eller patienten till följd av elektriska stötar, mekaniska faror eller andra faror.

OPOZORILO (SL)

Ta servisni priročnik je na voljo samo v angleškem jeziku.

- Če ponudnik storitve stranke potrebuje priročnik v drugem jeziku, mora stranka zagotoviti prevod.
- Ne poskušajte servisirati opreme, če tega priročnika niste v celoti prebrali in razumeli.
- Če tega opozorila ne upoštevate, se lahko zaradi električnega udara, mehanskih ali drugih nevarnosti poškoduje ponudnik storitev, operater ali bolnik.

DİKKAT (TR)

Bu servis kılavuzunun sadece ingilizcesi mevcuttur.

- Eğer müşteri teknisyeni bu kılavuzu ingilizce dışında bir başka lisandan talep ederse, bunu tercüme ettirmek müşteriye düşer.
- Servis kılavuzunu okuyup anlamadan ekipmanlara müdahale etmeyiniz.
- Bu uyarıya uyulmaması, elektrik, mekanik veya diğer tehlikelerden dolayı teknisyen, operatör veya hastanın yaralanmasına yol açabilir.

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Revision History

Revision	Date	Description
1.0	May 28, 2014	Initial release.
2.0	October 20, 2014	Updated manual per validation procedures. Updated parts return instructions. Added Section 1–9, Personnel Requirements. Added Run M3380, Hart ID cable, to cable map for connecting to either PED RBOX or dock cable and replacement of M1312. Added information on coiling excess cables. Added installation of terminators on UPM modules. Updated body coil tuning process flowchart.

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Table of Contents

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Chapter 1 Overview

1 Getting Started

1.1 Upgrade Introduction

This publication documents the mechanical installation information for upgrading an MR750 to MR750 with Dual Drive RF Amplifier.

WARNING

FERROUS MATERIAL HAZARD!
DELIVERY EQUIPMENT AND OTHER TOOLS AND PARTS REQUIRED FOR THIS INSTALLATION CONTAIN FERROUS MATERIAL AND WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.
KEEP ALL FERROUS TOOLS AT LEAST 10 FEET AWAY FROM THE MAGNET.

1.2 Site Ready Check

Pre-installation work must be completed before equipment is delivered to avoid delays and confusion.

Make sure that the proper hardware for installing the upgrade is present:

- RF cables in Equipment room must be either “Long” or “Short” length
- RF Cables in Magnet room must be either “Long” or “Short” length
- If needed, will Eco-power Upgrade be installed? Is hardware on site?
- If Eco-Power Upgrade is being installed and if site has MNS option, is the proper hardware on site?

1.3 Pre-Delivery FDO Checks

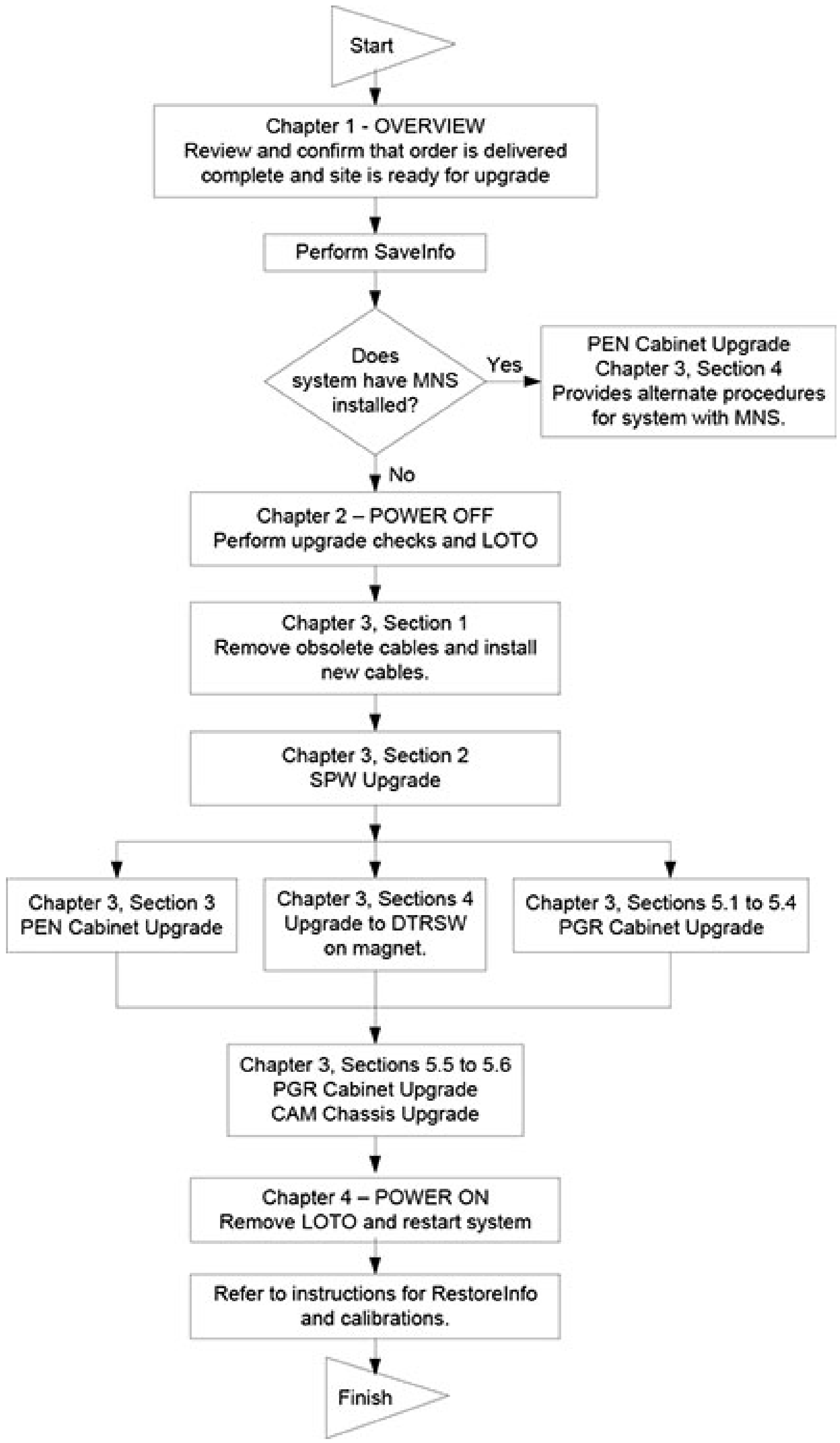
Before the shipment is made, Be sure that FDO review has been completed by consultation with the Upgrade Specialist. Also verify the Surface coil compatibility issues have also been resolved.

1.4 Upgrade Schedule Considerations

FMIs, periodic maintenance, installation of prerequisite upgrades and/or options will directly add to the man-hour and system down-time total. Be sure that all planned activity is considered when predicting the return of system to customer. Also verify Application Specialist is scheduled at the end of the installation to instruct users. Additional time must also be scheduled to instruct users following completion of upgrade.

1.5 Upgrade Flowchart

Illustration 1-1: Upgrade Flowchart



1.6 Product Delivery Instructions and Packing

The "Shipping Document" lists catalog numbers delivered. **Review and confirm that order is delivered complete.** Check impact on installation schedule if Catalogs and/or packing boxes are missing and/or noted as shipped short.

Labels that are attached to the outside of packing boxes summarize box contents. PDI (Product Delivery Instructions) specify box contents, part numbers, and shipping procedure. The PDI is numbered according to the catalog number. Lists of items included with each box are detailed by separate checklists, or a separate sheet that provides a summary of that box contents. Refer to PDI and packing lists for information specific to your shipment.

A set of service and operator manuals is delivered with the Upgrade Kit. Refer to checklists packed with "Technical Publication" boxes for a list of delivered documents.

1.7 Damage in Transportation

All packages should be closely examined at time of delivery. If damage is apparent, have notation **"damage in shipment"** written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period. File a report with:

- Call 1-800-548-3366. Select "Install Support Services for FOA and MIS"
- Contact your local service coordinator for more information on this process.

1.8 Product Locator

The Global Install Base Database (also known as Product Locator System) tracks shipment, trans-shipment, and field location of the serialized models. There are now two methods for submitting Product Locator information.

1. At this time, for **U.S. ONLY**, the preferred method of submitting information is the FE Site Verification Web Site at:

   http://gein2.med.ge.com/gib

   The FE Site Verification consists of three components that are available on the web from the main menu. They are:

   - Install/deinstall product locator model and serial numbers
   - Add/modify ship to address information
   - Update CARES FE data for primary/secondary FEs

   To obtain a copy of the FE training tutorial for using this website, a downloadable copy is available at the following: Product Locator Support Central Page - http://supportcentral.ge.com/15563

2. One "Shipping Card" is filled out and submitted when shipped (extra cards are supplied for trans-shipments between storage and distribution points), and the "Installation Card" and extra shipment cards are attached.

Verify that serial and model number on each rating plate matches installation card numbers before removing installation card. Note that there may be one or more shipment cards and bar code labels with the installation card. These shipment cards are used to trace the transfer of serialized units between various inventory storage and distribution points until the product reaches its final installation destination. Process just the installation card and discard any extra shipment cards and labels.

1.9 Personnel Requirements

Two people are required for this upgrade with at least one person trained in body coil tuning. This upgrade can take up to six days to complete.

1.10 Required Equipment and Tools

NOTICE

Follow all required safety and PPE procedures customary for your organization when working on this product.

Table 1-1: General Safety Requirements

Item	Description
1	Wear composite safety shoes for foot protection.
2	Wear gloves for hand protection.
3	Wear safety glasses for eye protection.

DANGER

FERROUS MATERIAL HAZARD!
IF MAGNET IS ENERGIZED, TOOLS, COMPONENTS, AND PARTS REQUIRED FOR THIS INSTALLATION THAT CONTAIN FERROUS MATERIAL WILL BE STRONGLY ATTRACTED TO MAGNET AND MAY BECOME DANGEROUS PROJECTILES.
KEEP ALL FERROUS TOOLS AS FAR FROM THE MAGNET AS POSSIBLE.

NOTICE

Only trained field service personal are permitted to use the RF Field Tuning Kits and are required to pass the training class in MyLearning, GBLSVCSEDTRNG-1549991, “ MR750 Dual Drive RF Field Tuning”.

In addition to the RF Field Tuning Kit, the RF Field Tuning software must be installed on a laptop computer prior to beginning the tuning procedure. The software can be ordered by a trained individual. Part number is (5352480-2).

DO NOT FIELD TUNE A BODY COIL UNLESS ABOVE TRAINING IS COMPLETED.

Tools to upgrade the system are listed in following table:

Table 1-2: Installation Equipment

Item	GE Part Number	Description
1		Screwdriver, slotted (long & narrow)
2	46-271138G1	Restricted Access Control Kit. Contains two plastic warning signs for posting at site during installation and service activity.
3		Service Methods DVD
4	46-198094P1	Wrist grounding strap
5	---	Plastic or aluminum flashlight
6	5307511–3 or (5434817–20 & 5434817 set)	RF Power Measurement Kit
7	46–194427P284	Digital Multi-meter (DMM)
8	5401455–2	Arc Detection Kit
9	5135527–2	Grafidy Kit
10	5311727	MR750 RF Field Tuning Kit
11	5336593–3	Network Analyzer
12		Center punch and hammer (if site has MNS option)
13		Round file, 1/4 inch (6mm) diameter (if site has MNS option)
14		All-in-one drill bit (if site has MNS option)
15		Power drill (if site has MNS option)
16		A small container made from a paper bowl, paper cup, heavy paper toweling, cloth, or coffee filter to catch metal filings if a hole needs to be drilled in the top of the PEN cabinet I/O panel. (if site has MNS option)

1.11 Discarded Material Return Policy

The equipment or material being removed by the upgrade have a potential value for service of the installed base. The disposition of such must be controlled by GE. Disposition of the removed material is per GE Medical Systems Policy and Procedures.

1.11.1 Material Policy Overview

1.11.1.1 Returned Material

Per GE Healthcare Policy 3.9: all requested material is to be returned, as directed, intact and promptly to Renewable Resources. Do not remove or swap any parts from the material to be returned.

1.11.1.2 Disposed Material

Regulations governing the disposal of this equipment or material vary from location to location and from country to country. In order to insure compliance with all local, state, or national environment, health, and safety regulations, dispose of all items per local GEMS policy and procedures and other applicable published guidance. Contact GEMS Recycling Center for assistance.

1.11.2 Procedure for Returning Material

All deinstalled UPM processor boards must be returned for recycling.

- Mark each box: MR750 Dual Drive Upgrade
- Send the boxes to the applicable region address in Section 1.11.5

1.11.3 Return Kit

If needed, packaging and/or transportation details can be obtained through the local Installation Specialist.

Reuse appropriate containers that new hardware and modules were shipped to site for return of above material.

1.11.4 Deinstallation Procedures

Refer to instructions in this manual for deinstallation procedures of site.

1.11.5 Region Specific Return Information

1.11.5.1 Sites in Americas

All deinstalled UPM processor boards must be returned to GE Renewable Resources for recycling:

```
GE Renewable Resources
Attn: Asset Recovery Leader, Harvest
7624 South 10th Street
Oak Creek, WI 53154
```

1.11.5.2 Sites in Asia

Return procedure for GEMS–Asia countries, contact:

```
GE HEALTHCARE JAPAN CORPORATION
ATTN: SWaP/HARVEST
67-4, Takakura-machi,Hachioji-shi-Tokyo
192-0033 Japan
Contact at Destination: 81-426-48-2595
FAX: 81-426-48-2902
```

1.11.5.3 Sites in Europe

For non-Gold Seal returns in Europe:

```
European Renewable Resources Center
GEODIS LOGISTICS - CD 26 - Pièce de la Remise - Bâtiment EVL2
Quai 45 à 49 - 91090 LISSES - FRANCE
TEL = (+33)-1-64-97-55-41 FAX = (+33)-1-60-86-12-31
```

1.11.6 Product Locator Information

Process product locator information for all serialized components removed from the system according to policy. Refer to Section 1.8.

1.12 Upgrade Components

The following is an overview of components required for upgrading a MR750 system to MR750 with Dual Drive.

-

3T Dual Drive RF Amplifier

- PGR Cabinet I/O Panel
- UPM Quiet Processor Boards
- Cables within PGR Cabinet
- Cables with Run numbers connecting between cabinets and components
- 3T Narrow Band Dual Drive Exciter and RF Q cable in PEN Cabinet
- 3T DTRSW with mounting brackets

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Chapter 2 Power Off Procedures

1 MR750 Dual Drive Upgrade Checks

It is recommended to obtain a baseline prior to starting the upgrade by running White Pixels and SPT.

Refer to the applicable MR Service Methods DVD to complete the following procedures.

1. Verify that all required upgrades have arrived complete and site is ready to shut down.
2. Do check scan to verify proper system operation – see service methods Check Scan (Good-Bye)
3. To confirm the UPM Processor boards are ready to harvest, run the following two UPM diagnostics. If the diagnostic completes, the UPM Processor boards are good to return. Fill in the attached Upgrade Test Procedure page located in the **Chapter 6, Appendix** of this manual. Make sure the test procedure is dated, signed, and all the * entries are completed.
    - Analog Data
    - I/O Data

    NOTE: The diagnostic location is:
    - Diagnostics >> System Function >> RF >> UPM Diagnostics

      OR
    - Diagnostics >> Hardware Location >> PGR Cabinet >> CAM >> UPM Diagnostics
4. Perform Save Info process – see service methods Save and Restore Tool

2 LOTO for PGR Cabinet

The Main Disconnect Panel (MDP) provides power to the HEC Cabinet, Cryogen Cooler Compressor, and the PGR Cabinet.

NOTICE

The following process only removes power from the PGR Cabinet. The HEC will continue to receive power so that the cryogen compressor can continue operation uninterrupted. Powering off the HEC circuit breaker will cause the Cryogen Cooler Compressor to also lose power. If this is done make sure that the HEC downtime is minimized to avoid boiling off Cryogens.

2.1 Procedure Overview

The Power, Gradient, RF Cabinet (PGR)Power Distribution Unit (PDU) and eXtreme Gradient Amplifier (XGA) reside in the PGR cabinet. The XGA is part of the eXtreme Gradient Driver (XGD) subsystem. Prior to servicing and/or maintenance work on the PGR PDU or any element of the XGD, the steps in this Lock Out / Tag Out (LOTO) procedure must be performed by a LOTO-authorized GE Field Engineer. Completing all the steps in this procedure ensures a safe environment and avoids equipment damage when working on these parts.

2.2 Components Affected by LOTO

Table 2-1: Affected by LOTO

Name of Equipment	Number of Locks	Titles of Employees Authorized to perform LOTO	Titles of Affected Employees	How to Notify
• eXtreme Gradient Amplifier (XGA or XG2) • eXtreme Power Supply (XPS or XP2) • Power, Gradient, RF Cabinet Power Distribution Unit (PGR PDU) • Gradient Filter	2 per GE Field Engineer (FE)	GE Field Engineers	Hospital Personnel	Verbal, Posted Signs

NOTE: The following table describes the type, location, and magnitude of energy to be LOTO'd.

Energy Source	Yes	No	Location of Energy Isolating Means	Magnitude of Energy
Electrical	x		PGR PDU Circuit Breaker Panel, Main Disconnect Panel (MDP) circuit Breaker Panel	208 VAC, 420 VAC, 380-480 VAC
Pneumatic		x		
Hydraulic		x		
Gas/Water/Steam		x		
Chemical		x		
Mechanical Motion		x		
Gravity		x		

Energy Source	Yes	No	Location of Energy Isolating Means	Magnitude of Energy
Springs		x		
Thermal		x		
Stored Energy	x		Time discharge of internal capacitors	208 VAC, 420 VAC
Air Under Pressure		x		
Oil Under Pressure		x		
Water Under Pressure	x		Heat Exchanger Cabinet (HEC), PGR Pump Circuit Breaker (PPMPCB) and RF amp quick disconnect	5.0 Bar (80 psi)
Gas Under Pressure		x		
Steam		x		
Other		x		

2.3 Preliminary Requirements

Table 2-2: Tools and Test Equipment

Item	Part #	Quanitity
Personal Lock	46-194427P320	2
Multi-locking Device (If multiple service people)	46-194427P313	1
Red Warning LOTO Tag	46-194427P322	2
Digital Voltmeter	46-194427P284	1

2.4 Safety

DANGER

ELECTROCUTION HAZARD!
HIGH VOLTAGE PRESENT.
USE PROPER LOCK OUT / TAG OUT PROCEDURES BEFORE INSTALLING, SERVICING AND/OR MAINTAINING.

2.5 Applying LOTO to PGR

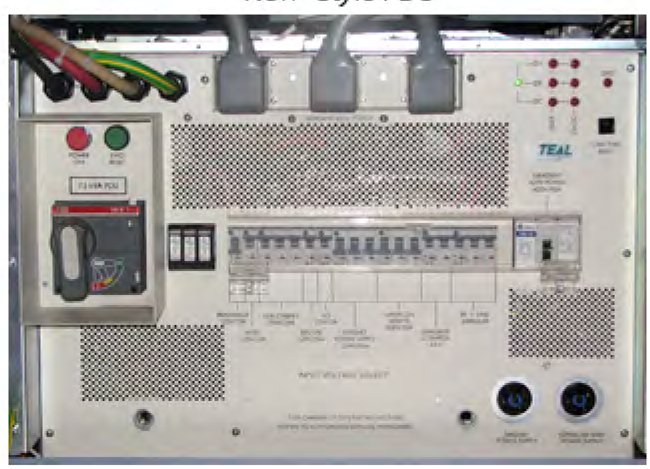
NOTE: There are two types of MDPs and two types of PDUs. The LOTO steps are the same, but the physical components are slightly different.

"Old" Style MDP

"New" Style MDP

"Old" Style PDU

"New" Style PDU



2.5.1 Types of Equipment and/or Methods Selected to Dissipate or Isolate Stored Energy

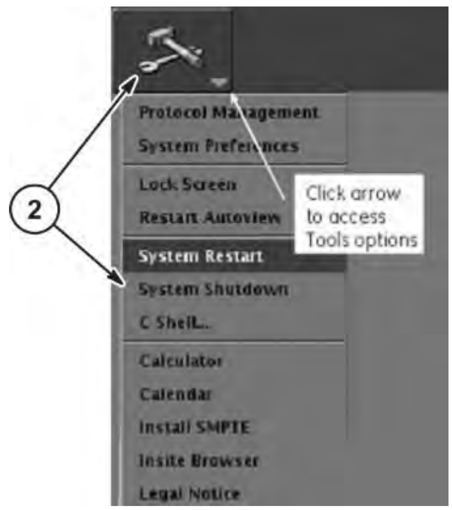
Time method for dissipation of stored energy – five minutes.

2.5.2 Type(s) of Equipment and/or Method(s) Used to Ensure Disconnection(s)

1. Two personal locks and tags per Field Engineer (FE) at the site working on the de-energized equipment at the time of LOTO.
2. Voltmeter to read voltage for verification that energy has dissipated in the PDU.

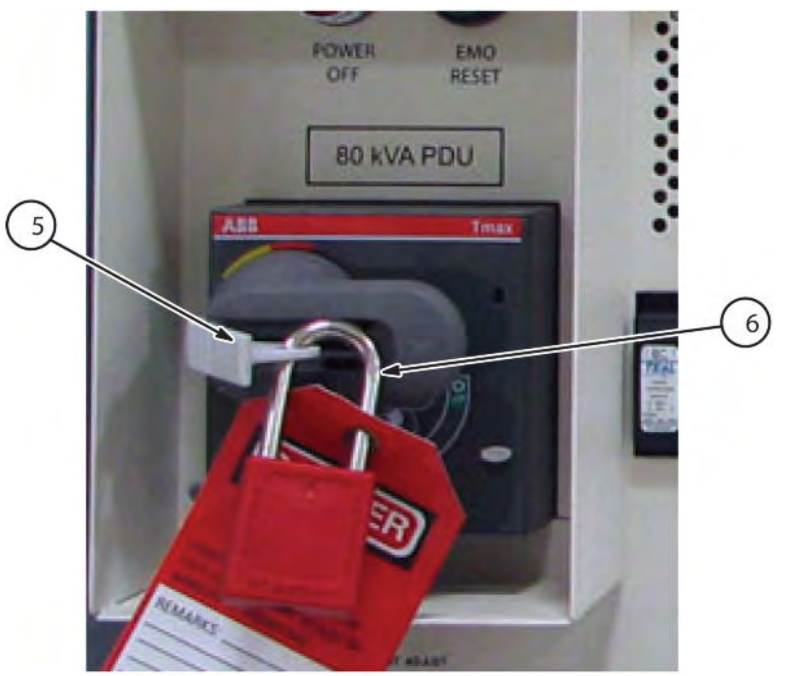
2.5.3 Applying LOTO

1. To prepare for shutdown of equipment, notify affected personnel working in the area the LOTO is being performed.
2. At the Operator Workspace, shut down the Host PC by clicking the arrow on the *Tools High Level Access* tab, select System Shutdown, and click [Yes] to confirm the shut down. Shutting down the Host PC shuts down the system safely and turns off the power to the Host PC

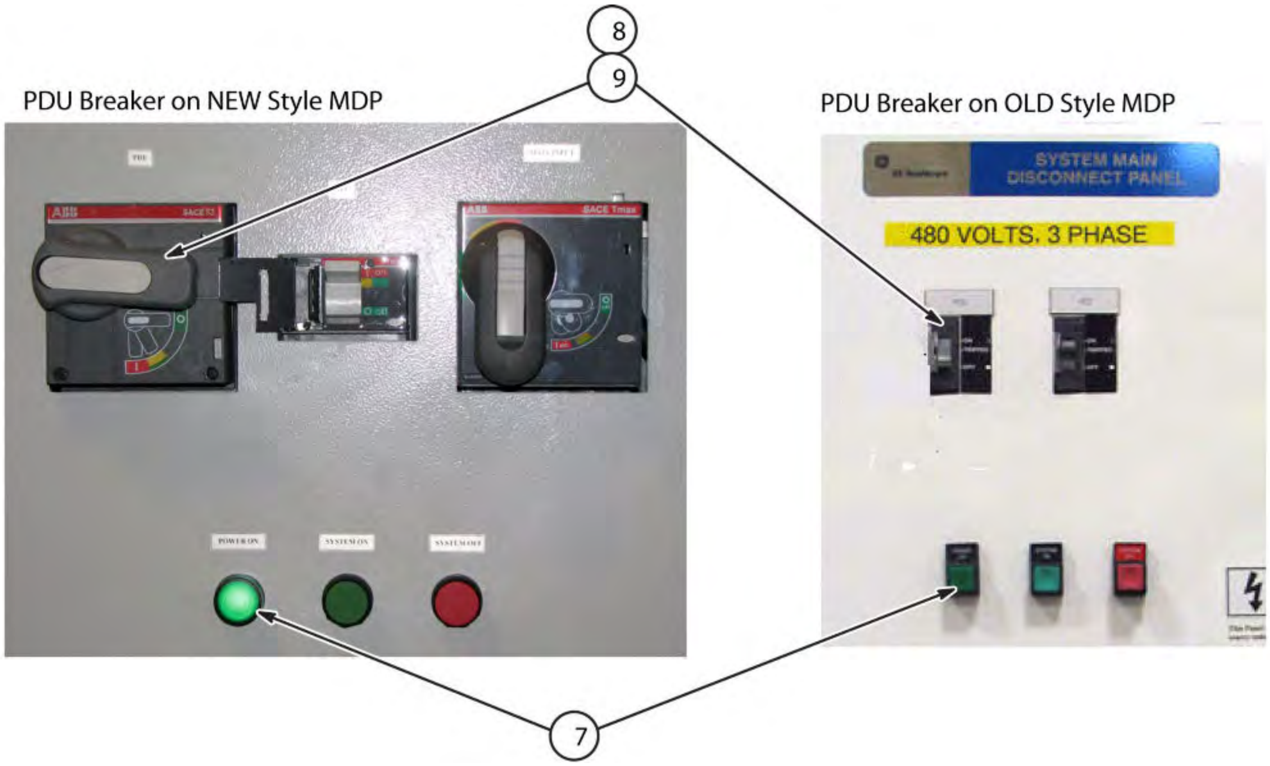


3. At the PGR cabinet, go to the front panel of the PGR PDU and press the POWER OFF button.
4. At the front panel of the PGR PDU rotate the large main breaker counterclockwise to the *Green O (OFF)* position.

5. At the front panel of the PGR PDU, pull out the locking mechanism on the end of the main breaker (or, on the new breaker, press on the arrow) to expose the hole where the Lock and Tag can be placed.
6. Place the LOTO lock on the PGR PDU main breaker.



7. Observe that the POWER ON light is on. It signifies the main breaker is closed and control power is on.
8. Locate and turn off the PGR PDU main breaker on the Main Disconnect Panel (MDP) that feeds the PGR cabinet.
9. Lock Out/Tag Out the PDU input switch on the MDP



10. Verify that all energy has been dissipated. Use a voltmeter to measure the line-to-line voltage between each **INPUT** phase voltage test point on the PDU. (Shown below.)

    NOTE: The test points are 100:1 ratio (480 VAC = 4.80 VAC, 208 VAC = 2.08 VAC). Therefore, a low voltage reading still indicates the presence of high voltage. **Your voltage reading for each of the three phases must be 0 VAC before proceeding.**

    Input Voltage Range: 380 - 480 VAC = 3.8 - 4.8 VAC

    Output Voltage Range: 208 VAC = 2.08 VAC

Illustration 2-1: PDU Test Points (PDU 5140621)

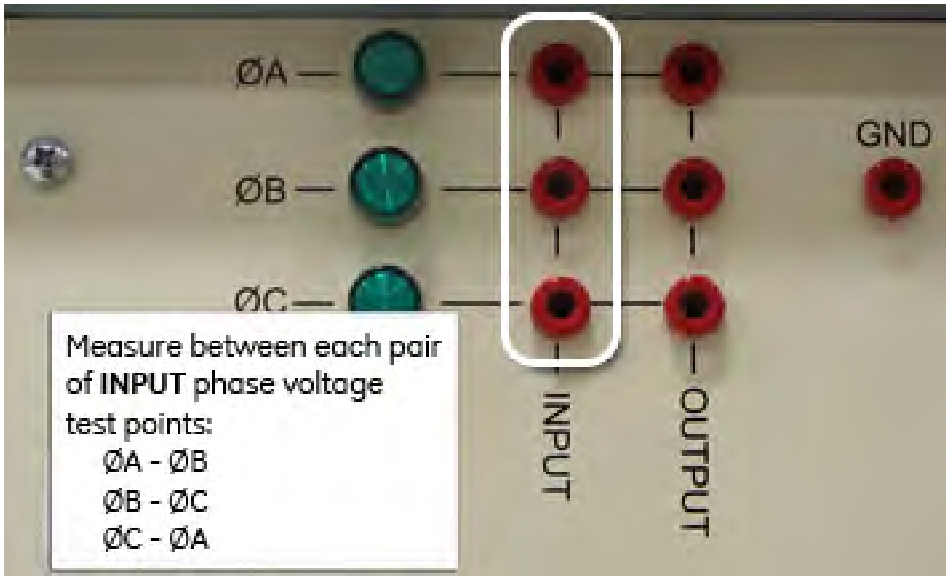
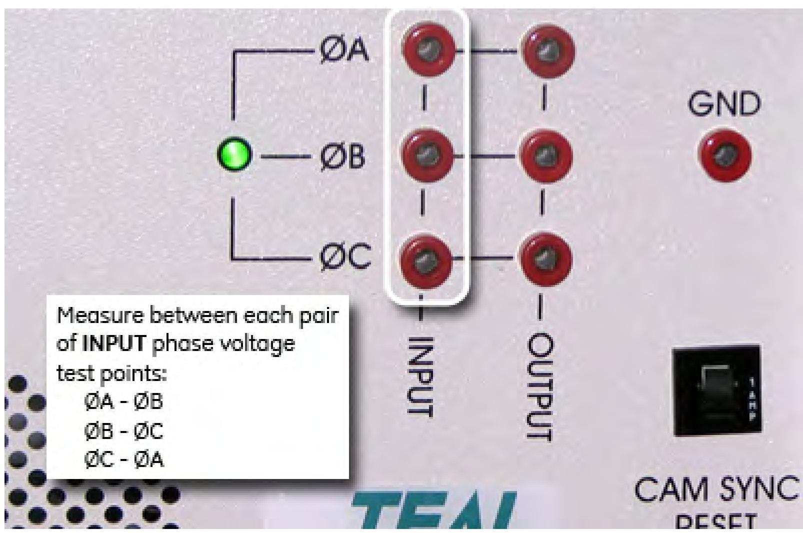


Illustration 2-2: PDU Test Points (PDU 5343114)



Chapter 3 Upgrade Procedures

1 Cable Interconnects Upgrade

NOTICE

This upgrade consists of the removal and installation of RF cables in the overhead tray. The MR Service Platform, part 5155291, must be used with the safety rails installed.

NOTICE

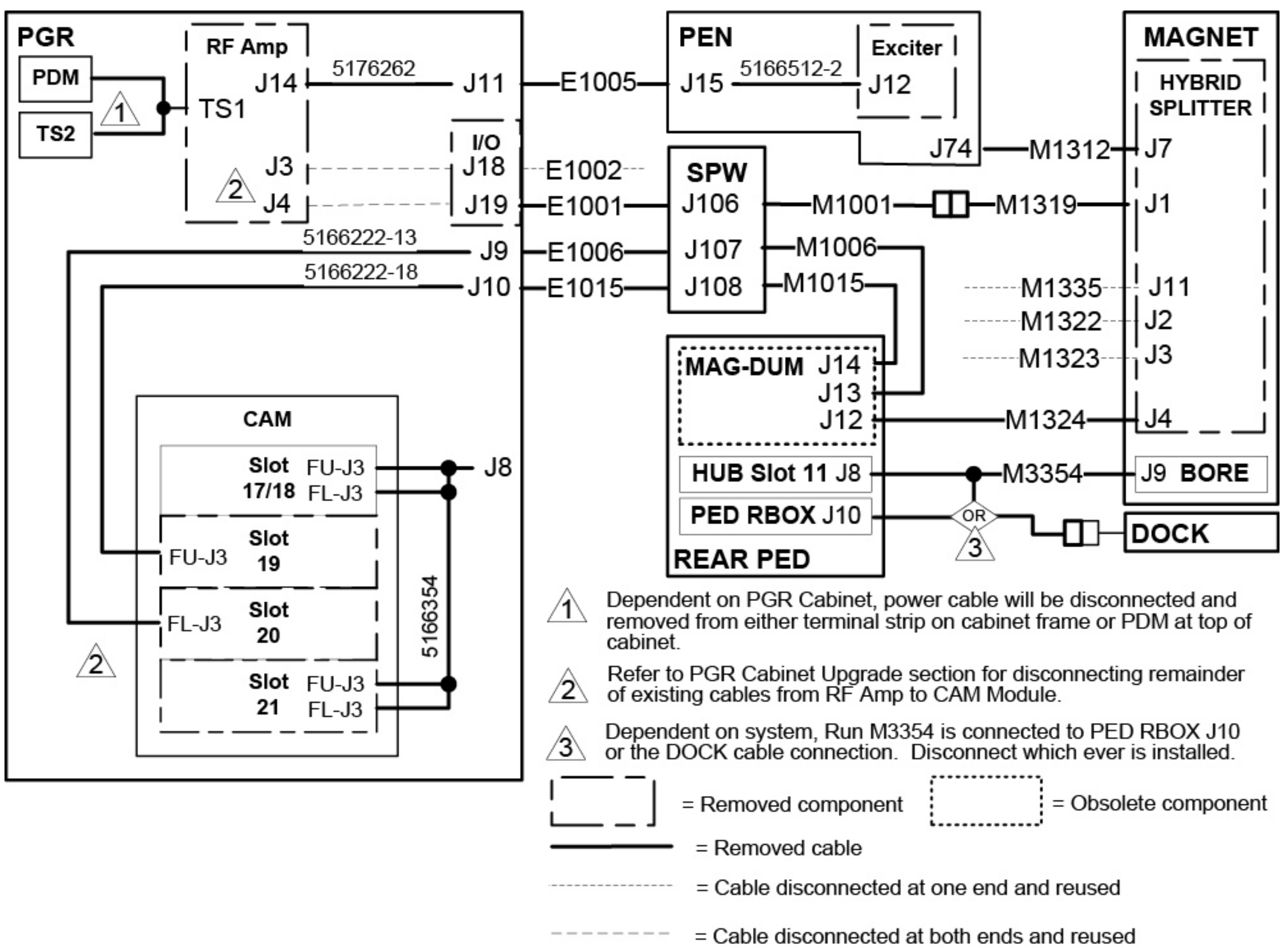
All existing RF Transmit cables MUST be removed when performing this upgrade to ensure proper image quality, and successful Dual Drive Quadrature calibration. Failure to comply will result in additional troubleshooting. the RF cables. The cables to be removed are routed from the Exciter, to the RF Amp, to the SPW, to the Hybrid Splitter, and then to the Body Coil.

This upgrade will remove and replace specific cables and also add new cables. The details for the upgrade procedures are found in the sections that follow.

1.1 Cable and Component Removal

The following illustration not only shows the cables and components that will be removed, it also shows the cables that will be disconnected and reused.

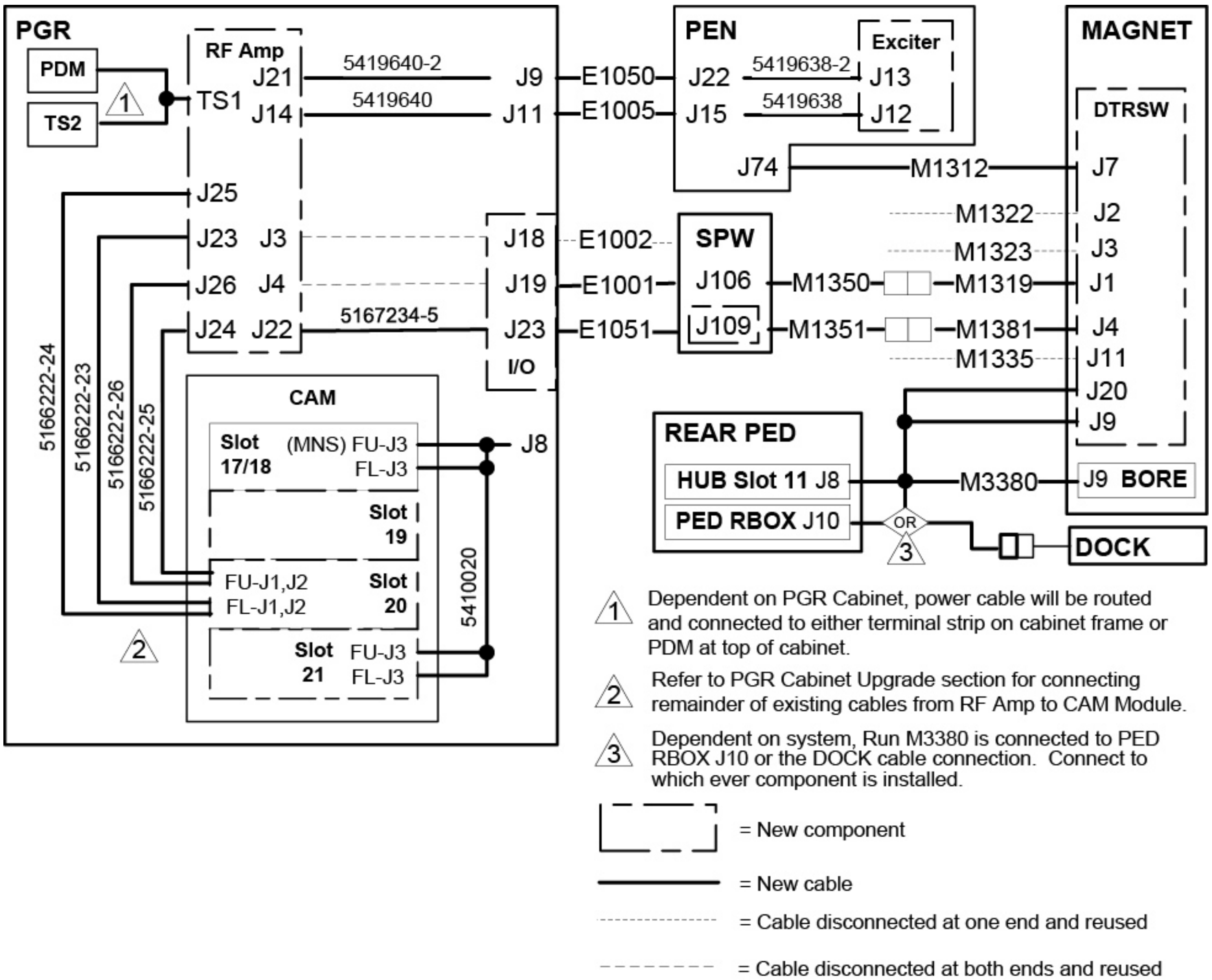
Illustration 3-1: Cable and Component Removal



1.2 New Cable and Component Installation

The illustration shows new cables to be routed and connected, existing cables to be reused, and new components to be installed.

Illustration 3-2: Cable and Component Installation

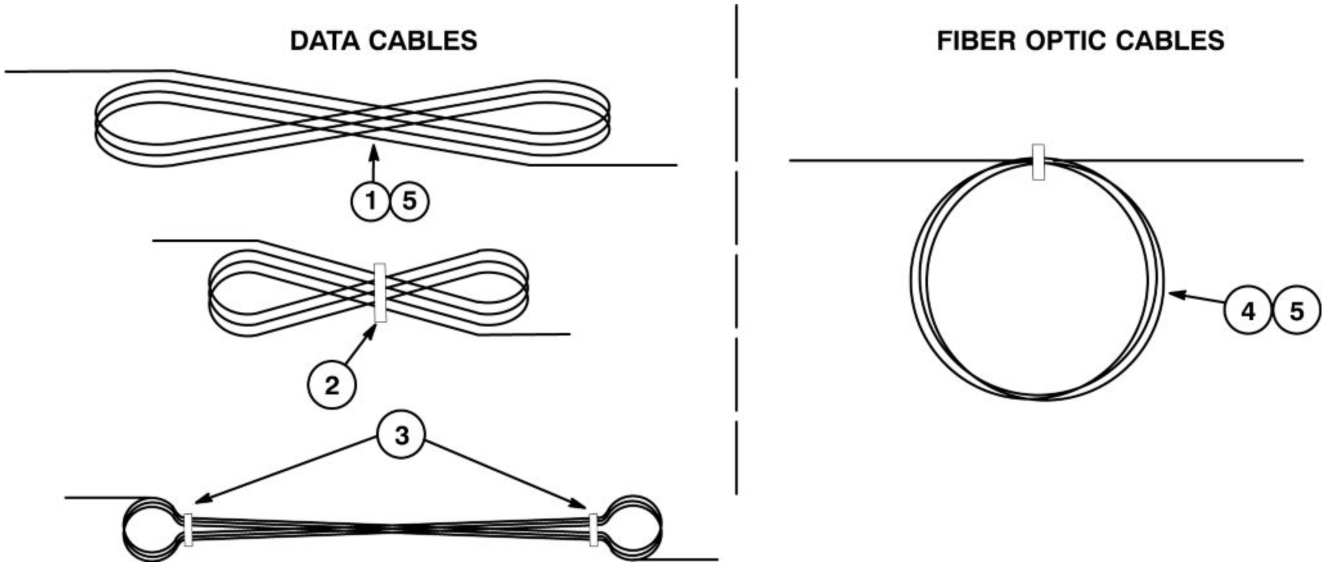


1.3 Coiling Excess Cable Length

The type of coiling used to store excess length of cables depends on the type of cable. Data cables are coiled differently than fiber optic cables..

- DATA CABLES - Excess length of cables must be coiled in a “Figure Eight” pattern. DO NOT COIL IN CIRCLES!
- FIBER OPTIC CABLES - Excess length of cables must be coiled in a circle pattern. DO NOT COIL IN A “FIGURE EIGHT” PATTERN!

Illustration 3-3: Proper Coiling of Excess Cables



1. Excess data cable to be stored in overhead cable trays or Penetration Panel closet must be routed in a "Figure Eight" pattern as shown.

NOTICE

The minimum bend radius of any cable must not be violated no matter how the excess cable is coiled.

2. For smaller figure eight patterns, dress and tie-wrap middle of looped cables.
3. For longer figure eight patterns, dress and tie-wrap ends of looped cables
4. Excess fiber optic cable to be stored in overhead cable trays, Penetration Panel closet, or in magnet enclosure area must be routed in a circular pattern as shown.
5. Refer to *Cables Installed on Site* table in the *Table Specification* Chapter of Direction 5500102, *Discovery MR450 & MR750 System Installation* manual which should be found on site
6. Repeat as necessary for cables in Magnet or Equipment room.

2 SPW Upgrade

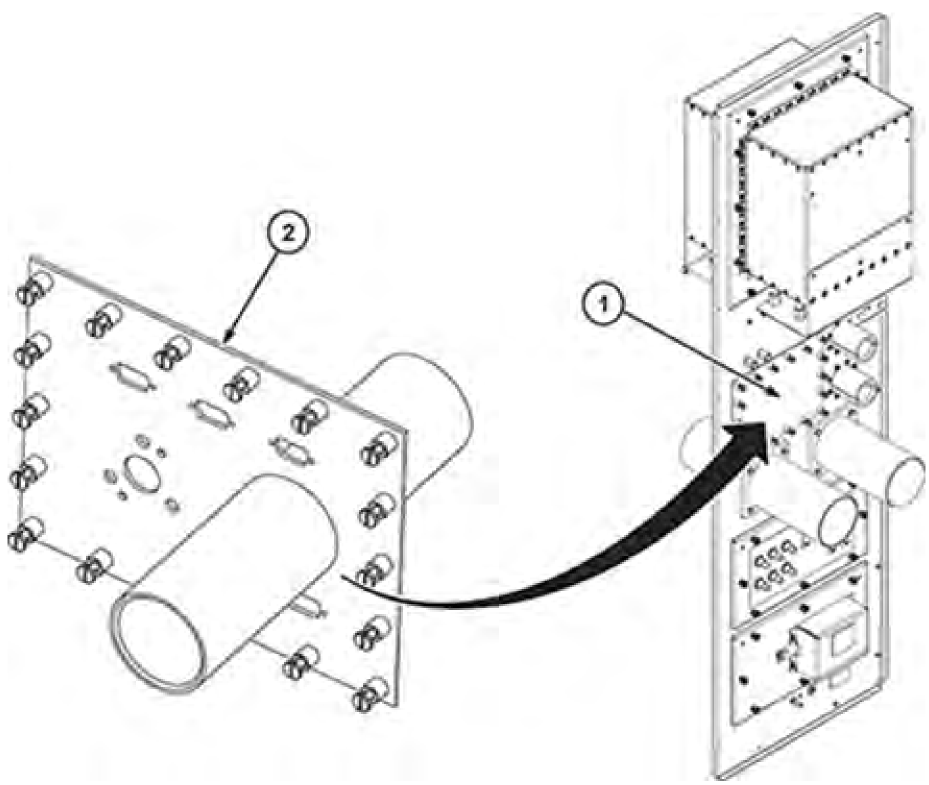
2.1 Disconnect Cables

1. On magnet room side of the SPW, disconnect and remove the following cables routed from the SPW to the bottom of the Magnet Dummy Load module at the back of the Rear Pedestal:
   a. Run M1001 from J106
   b. Run M1006 from J107
   c. Run M1015 from J108
2. On the equipment room side of the SPW, disconnect and remove the following cables routed from the SPW to the top of the PGR Cabinet:
   a. Run E1001 from J106
   b. Run E1006 from J107
   c. Run E1015 from J108

2.2 Install Dual Drive Panel

1. Remove existing panel on SPW and discard.
2. Install new Dual Drive Panel (2394034–20) for J109 connection.

Illustration 3-4: SPW Dual Drive Panel Installation



2.3 Connect New Cables

1. On magnet room side of the SPW:
    a. Route new cable Run M1350 from Run M1319 at the top of the magnet, through the overhead cable trough, and connect to J106 on SPW.
    b. Route new cable Run M1351 from Run M1381 at the top of the magnet, through the overhead cable trough, and connect to J109 on SPW.
2. On the equipment room side of the SPW:
    a. Route and connect new Run E1001 from the top of the PGR Cabinet, through the overhead cable trough, and connect to J106 on SPW.
    b. Route and connect new Run E1051 from the top of the PGR Cabinet, through the overhead cable trough, and connect to J1069 on SPW.

3 PEN Cabinet Upgrade

3.1 Overview

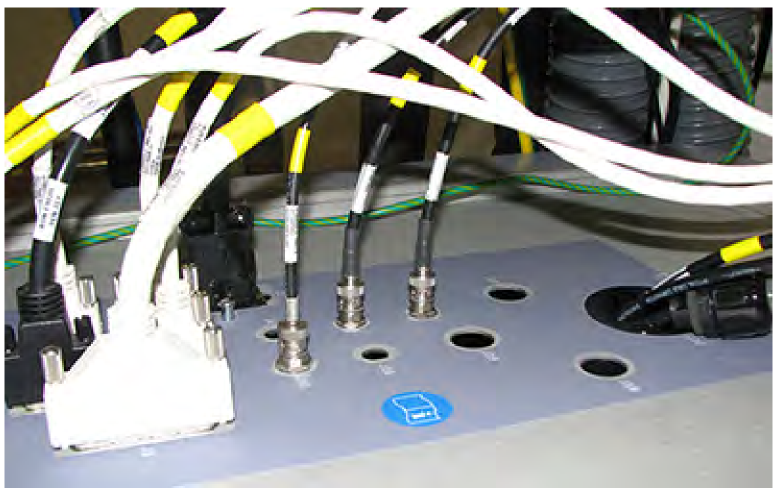
The existing Exciter module will be replaced by a new 3.0T Narrow-Band Dual-Drive Exciter module with the addition of a new cable and replacement of an existing cable.

3.2 Create J22 Connection Point in I/O Panel

A new TX cable (5419638–2) is to be installed and will be routed from J22 in the top I/O panel to J13 on the new Exciter module. To complete this cable installation, a J22 connection point needs to be located in the top I/O panel. Dependent on existing cables connected to the I/O panel, there are two possible methods to complete the new cable installation.

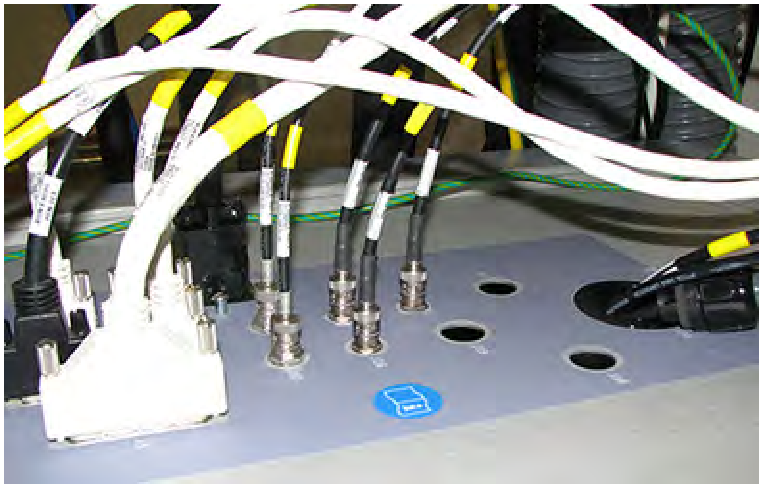
- If the system does not have MNS, then there are two BNC connection points on the I/O panel that are not used, J4 and J16. For this upgrade, J16 will be remarked as J22 and used for installing the new cable.

Illustration 3-5: I/O Panel without MNS



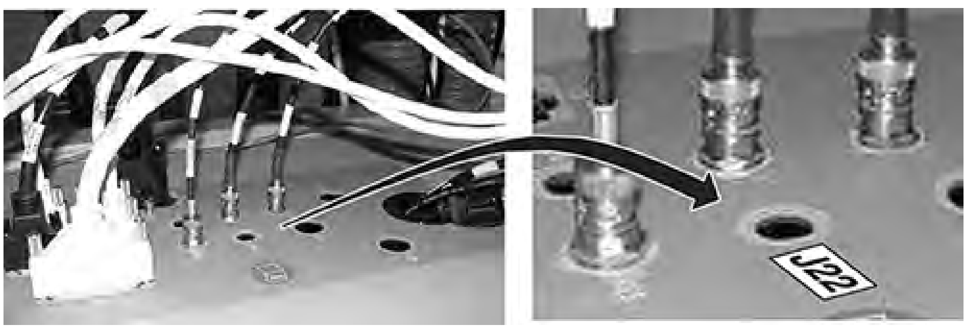
- If the system has MNS, then a new hole will have to be provided and marked as J22. Refer to Section 3.2.2 for instructions on creating this hole.

Illustration 3-6: I/O Panel with MNS



3.2.1 J22 Connection in System without MNS

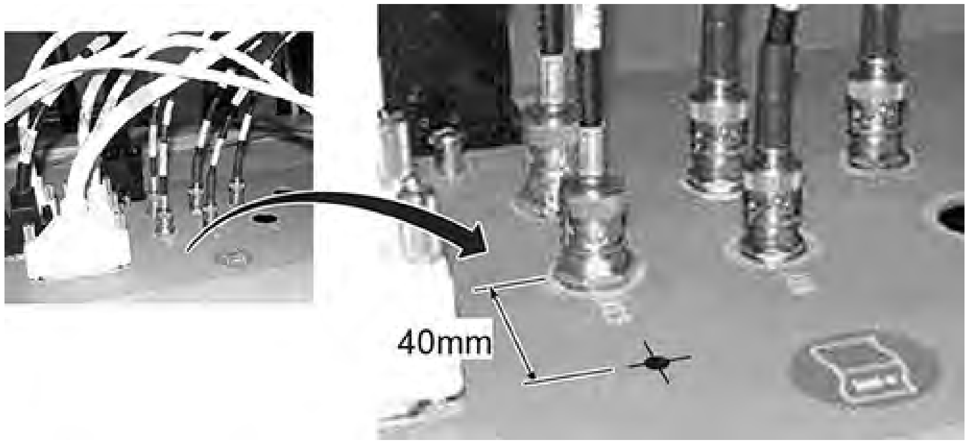
Illustration 3-7: I/O Panel J22 Connection without MNS



1. At location J16, apply a piece of masking tape over J16 and write J22 on the tape.
2. Install the BNC end of the TX cable from bottom side of I/O panel and secure with the supplied nut. Route the cable along the path of existing cables to the front of the Exciter module.

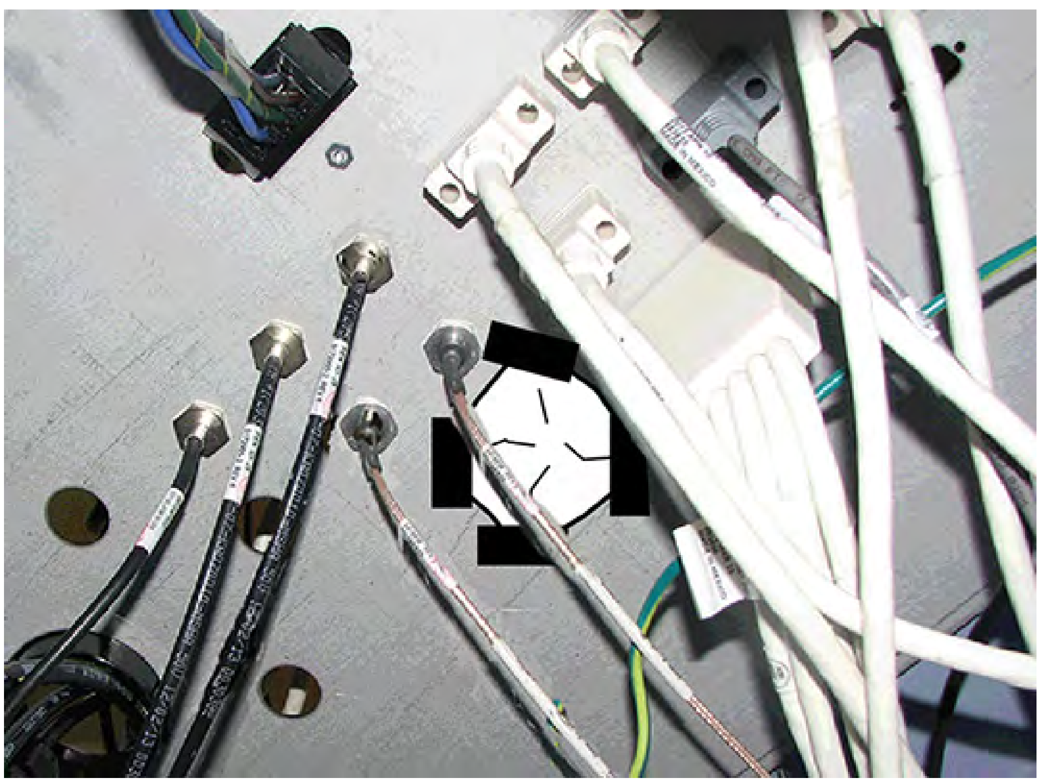
3.2.2 J22 Connection in System with MNS

Illustration 3-8: Locate New BNC Hole



1. Measure 40mm (1.6 inches) from the center of J15 and mark the location.

Illustration 3-9: Preparation for Drilling Hole



2. In preparation for creating the mounting hole for J22, a “diaper” has to be fabricated from a cloth or heavy paper towel, to catch any metal shavings that might fall as the hole is being drilled. Tape the diaper to the bottom of the I/O panel where the hole is to be drilled.

   NOTE: If needed, disconnect cables around area of hole to be drilled. This would provide a larger area for securing the diaper.

3. Using a 12mm (1/2 in) drill bit, drill the hole through the top of the I/O panel at the J22 mark .

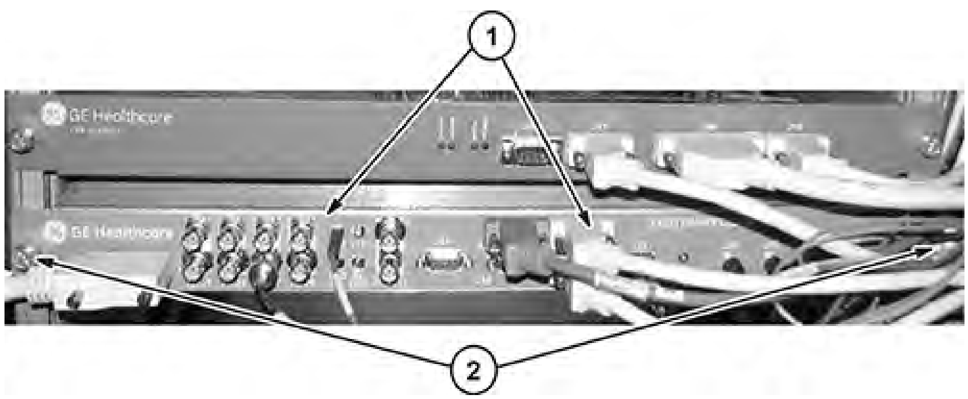
- Make sure the drill bit does not penetrate the diaper
- Make sure there are no sharp edges around the hole when complete. Use the round file to remove any burrs.
- Collect any filings from the top surface of the I/O panel

4. Carefully remove the diaper from the bottom side of the I/O panel and discard. Make sure that no metal filings fall onto any modules.
5. Apply a piece of masking tape next to the hole and write J22 on the tape.
6. Install the BNC end of the TX cable and secure with the supplied nut. Route the cable along the path of existing cables to the front of the Exciter module.

3.3 Replace Exciter Module

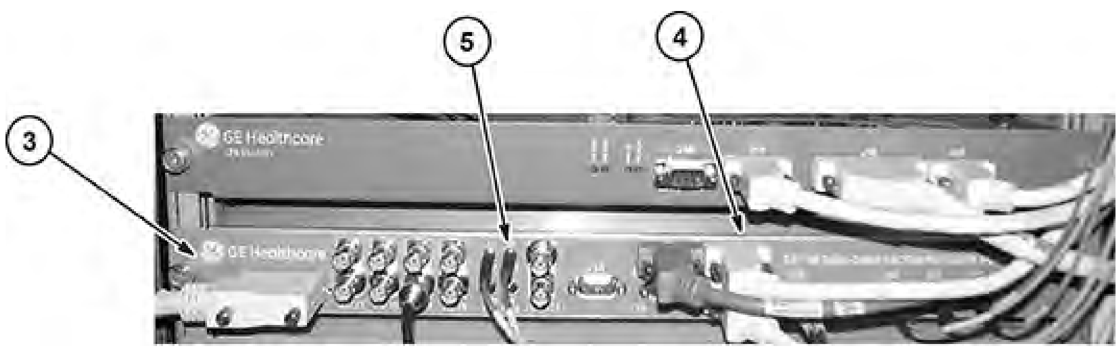
The existing Exciter module will be replaced by the new 3.0T Narrow-Band Dual-Drive Exciter module.

Illustration 3-10: Remove Existing Exciter Module



1. Disconnect all cables from existing Exciter module. Remove and discard cable (5166512–2) routed from J12 on Exciter module to J15 on I/O panel. A new cable will be installed to replace this one.
2. Unfasten screws and slide out module. Return for recycling.

Illustration 3-11: Install New Exciter Module and Cable



3. Unpackage and slide in new Dual Drive Exciter Module (5490144–2) and fasten with previously removed screws.

4. Connect all cables previously removed.
5. Connect new RF Tx cable (5419638–2) from J22 on I/O Panel to J13 on front of Exciter module. Route and connect the other Tx cable (5419638) from J15 on I/O Panel to J12 on Exciter.

4 Dual Drive Upgrade

This procedure replaces the existing Body Hybrid Splitter module with a new DTRSW Assembly (Dual Drive). The flanged bearings used to secure the side electronics plate will also be replaced with stainless steel washers.

4.1 Safety

⚠WARNING

SHOCK HAZARD!
HIGH BIAS VOLTAGES ARE PRESENT AT THE BODY HYBRID DURING NORMAL OPERATION.
REMOVE POWER AND PERFORM LOTO ON THE PDM-RF POWER ON PDU BEFORE ACCESSING THE BODY HYBRID. SEE THE “MR SERVICE SAFETY MANUAL,” PN 5452735.

⚠CAUTION

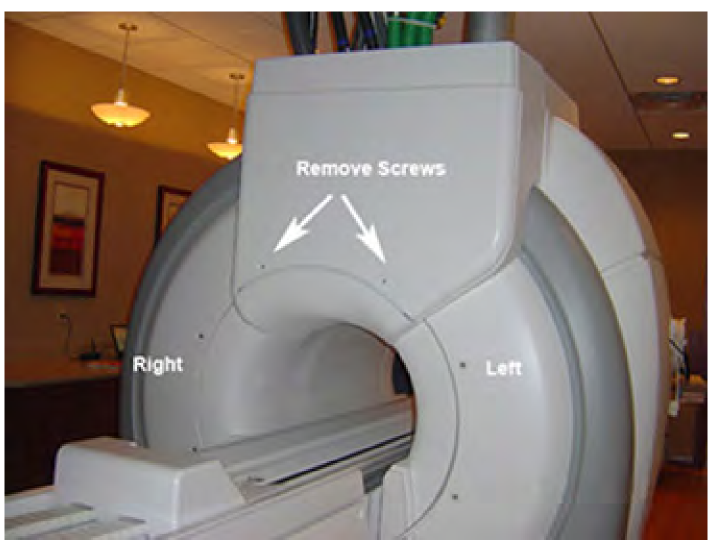
Strong Magnetic Field!
The module has no ferrous material and is not attracted to the magnet, but eddy currents set up by the strong magnetic field interacting with the body hybrid module can make it very hard for one person to handle.
When removing or replacing the body hybrid module from the side of the magnet, always do so with two people, keeping the module as far away from the magnet as possible.

4.2 Enclosure Cover Removal

1. Perform LOTO on the PDM-RF power for the system. See the MR Service Safety Manual, PN 5452735.

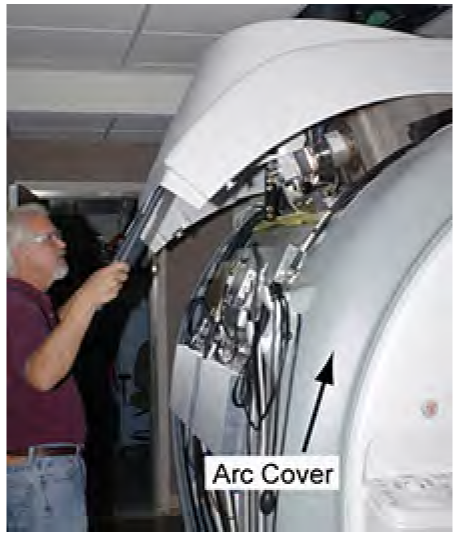
2. Make sure the scanner is idle. Select [End Exam] at any point in the scan process.

Illustration 3-12: Rear Cable Cover Screw Locations



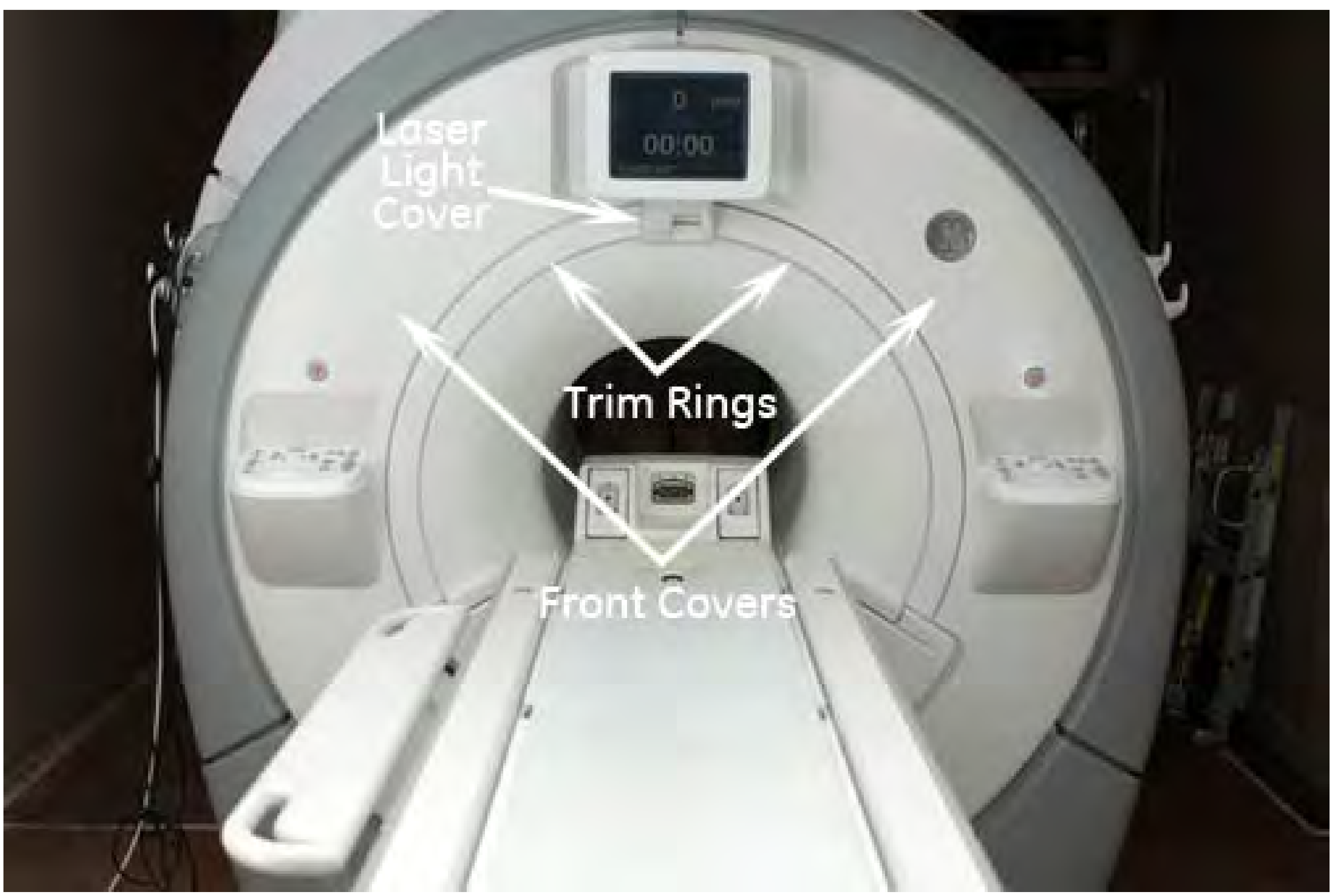
3. Remove the two Rear Cable Cover screws and slide the cable cover straight up to disengage from the upper hooks.

Illustration 3-13: Removing Upper Magnet Side Cover



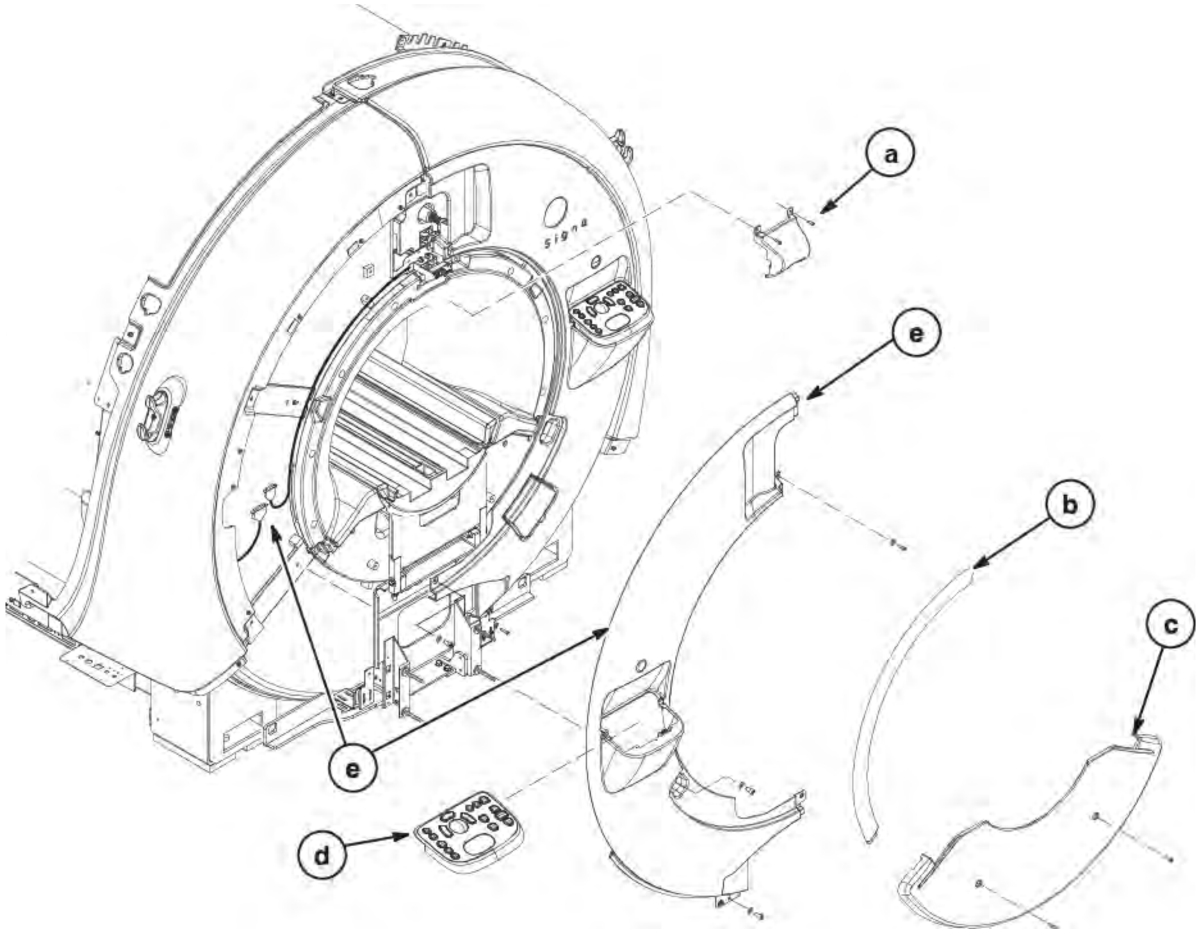
4. Remove the lower side cover and upper magnet side cover.

Illustration 3-14: Left Front Covers



5. Remove left side covers at front of magnet as follows:

Illustration 3-15: Remove Front Covers



a. Remove the alignment laser light cover by unscrewing two screws. The cover is held in place by a popper on the lower left corner and snaps off.

b. Remove the two trim rings by pulling away from the enclosure. They are held in place by poppers.

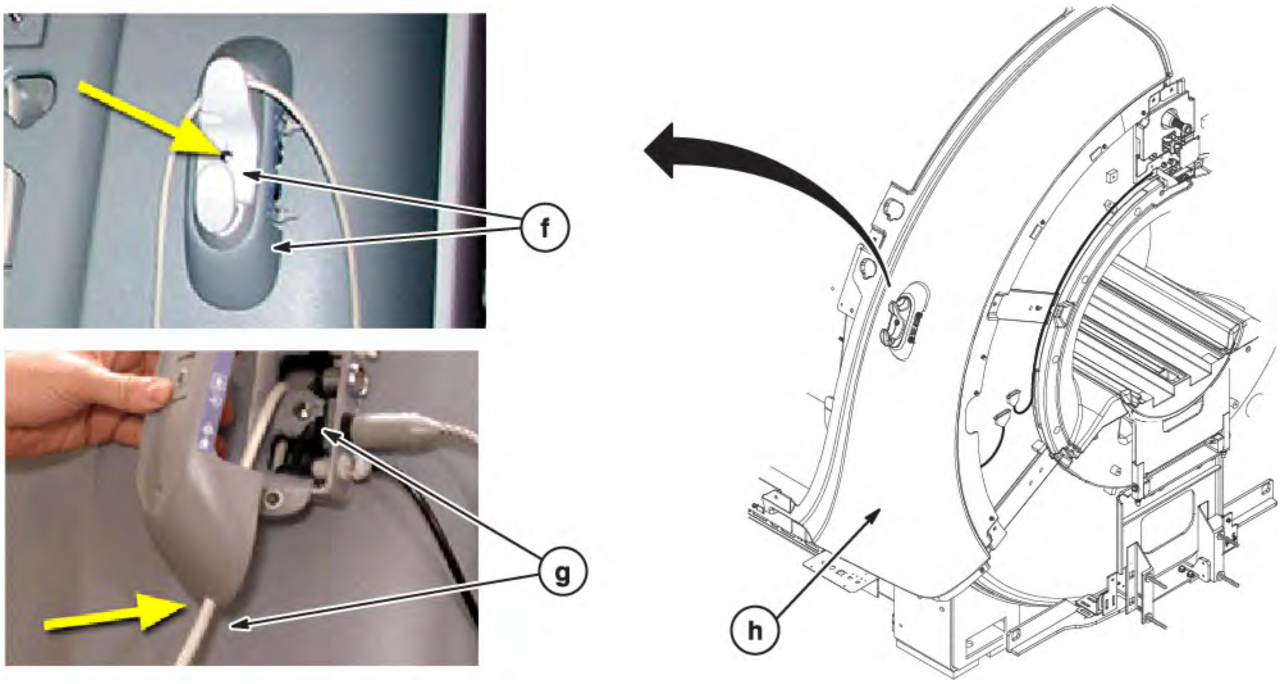
c. Remove the two M6 screws that secure the front bridge cover.

d. If necessary, remove the left control panel cover. The cover is also held on by poppers and Velcro.

NOTE: Be careful of the trackball control panel connection cables.

e. To free the left front cover, disconnect the two cables that connect the control panel to the magnet.

Illustration 3-16: Remove Remote PAC Assembly



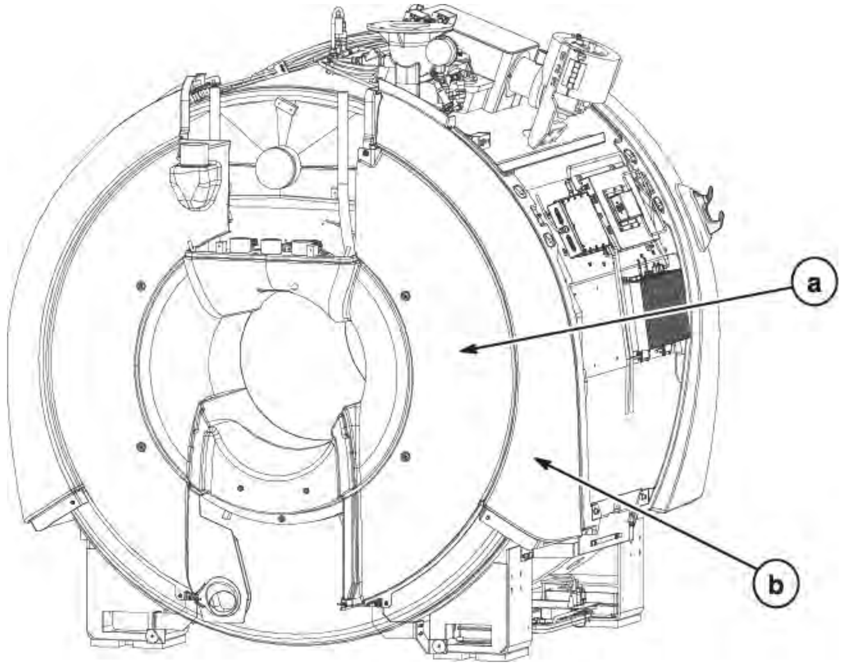
f. At the Remote PAC Assembly on the arc cover, remove all cables. Remove the pan head screw holding the white hook. then remove the two screws holding the grey base to the bracket.

g. The PPG cable, along with other cables are routed through the hole in the arc cover. Take care to not damge the cables when removing the arc cover in the next step.

h. Remove all the M10 screws attaching the arc cover toe the magnet. Move the arc cover far enough away to provide access to the two side electronic plate mounting bolts.

6. Remove left side covers at rear of magnet

Illustration 3-17: Remove Left Rear Covers



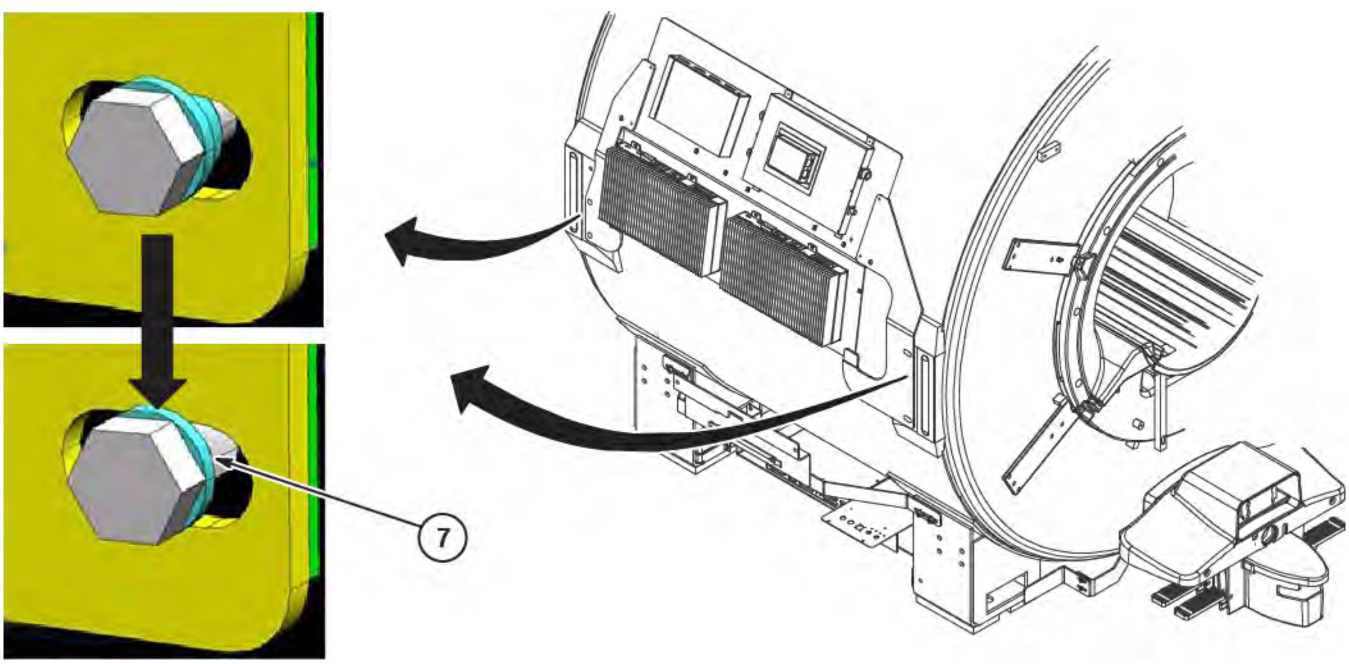
a. Unfasten button head screws and remove left rear cover.

b. Unfasten M10 screws and remove left rear arc cover.

4.3 Replace Side Electronic Plate Mounting Washers

Replace the non-metallic flanged bearing with M12 Stainless Steel Washer (2001–M12–10) at the four mounting locations

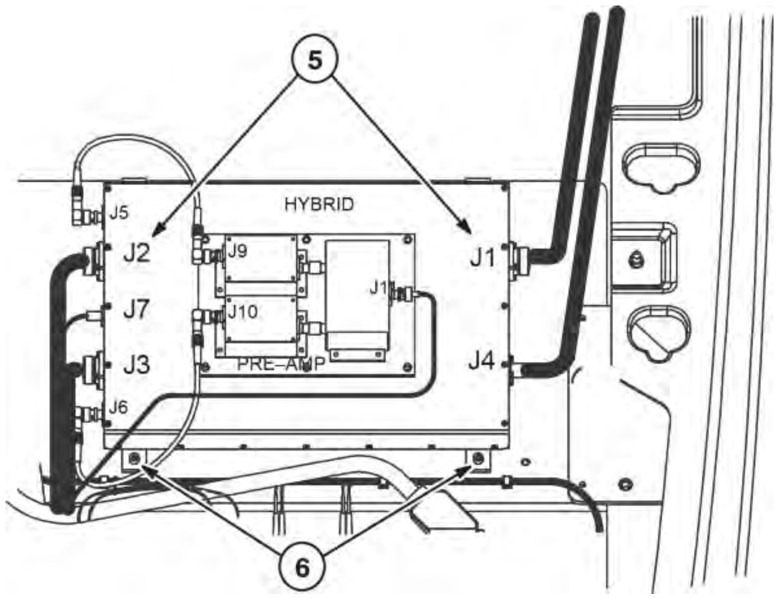
Illustration 3-18: Replace Flange Bearings with Washers



4.4 Dual Drive Installation

4.4.1 Remove Body Hybrid Splitter

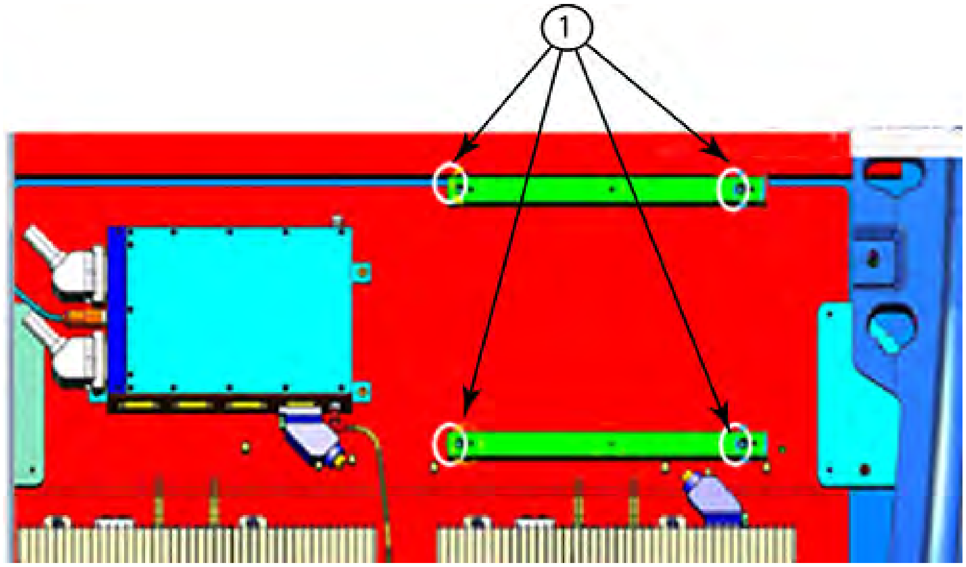
Illustration 3-19: Remove Body Hybrid Module



1. Disconnect cables connected to J1, J2, J3, J4, J7, and J11 from the body hybrid module.
2. Unfasten the four screws, two at the top and two at the bottom, and remove the body hybrid module.
3. Cable Runs M1312, M1319 and M1324 that were disconnected fromJ7, J1 and J4 are to be discarded and will be replaced by new cable Runs M1312, M1319 and M1381.
4. Package and return body hybrid module for recycling.

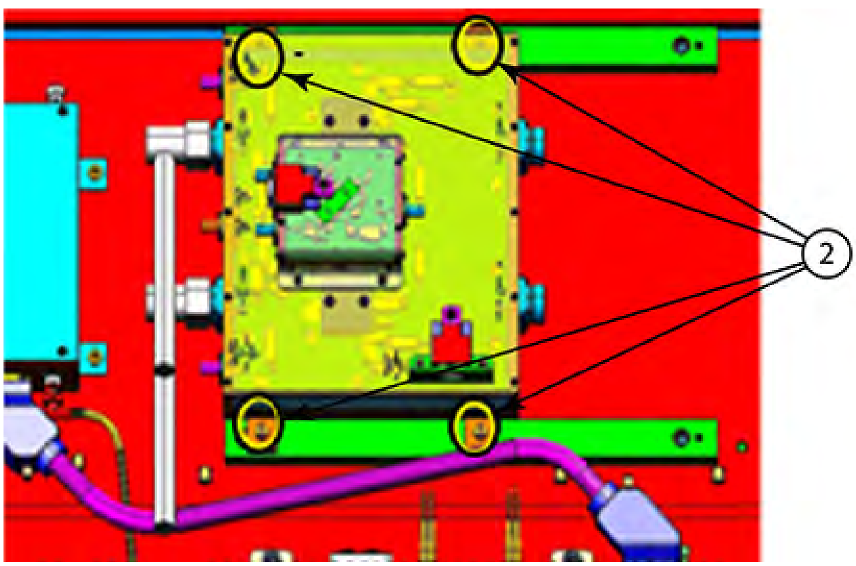
4.4.2 Install New DTRSW Module

Illustration 3-20: Attaching Mounting Brackets



1. Position the two mounting brackets (5439493) on side mounting plate and attache each with two M6 x 16mm longHex Socket Cap Screws (1000–M6C016–31).

Illustration 3-21: Install DTRSW Module



2. Attach DTRSW Module to brackets with the four M6 captive screws that are part of the module assembly.
3. Connect existing cables Runs M1322, M1323, and M1335 to J2, J3, and J11 as marked.
4. Connect new Run M1312 to J7. Route other end of cable to PEN Panel and connect to J74
5. Route and connect the two cable ends of M3380 marked J20 and J9 to DTRSW and pre-amp.

6. Connect NEW cable Runs M1319 and M1381 to J1 and J4 as marked and route towards the top of the magnet. They will be connected to Runs M1350 and M1351.

4.5 Reinstall Covers

Reverse the steps in to install front, rear, and side covers.

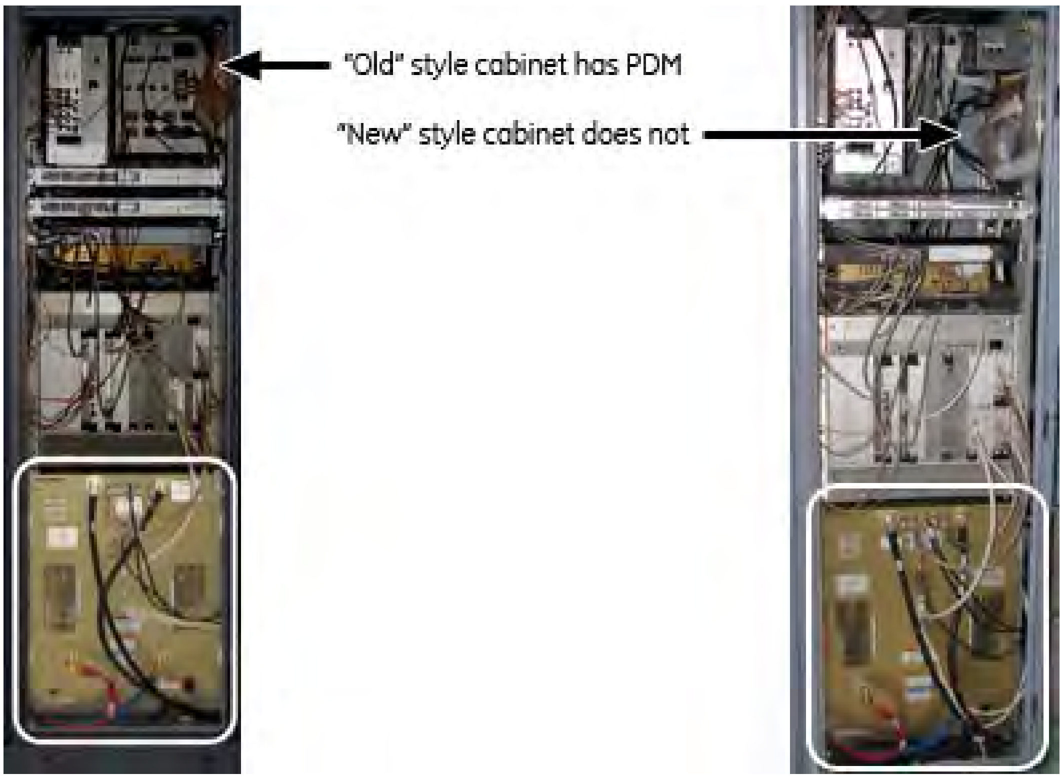
5 PGR Cabinet Upgrade

5.1 RF Amp Removal

There are two styles of PGR Cabinets. The “old” style contains a Power Distribution Module (PDM) in the upper right corner of the cabinet. The “new” style does not contain a PDM.

This section contains the procedure for the replacement RF Amp. the unit weighs about 315 pounds and a hoist kit is required. This document details the correct way to use a hoist kit to safely remove and the RF Amp.

Illustration 3-22: Two Styles of PGR Cabinet



WARNING

POSSIBLE PERSONAL INJURY OR EQUIPMENT DAMAGE!
THE MODULE COULD FALL WHEN USING THE LIFT TOOL.
BEFORE PUTTING WEIGHT ON THE HOIST CHAIN, BE SURE THAT THE LINKS ARE ALIGNED PROPERLY. THIS HELPS KEEP THE CHAIN FROM BINDING UP, OR BECOMING KINKED, WHICH IS NEARLY IMPOSSIBLE TO CORRECT WHEN THE CHAIN IS UNDER TENSION.

NOTICE

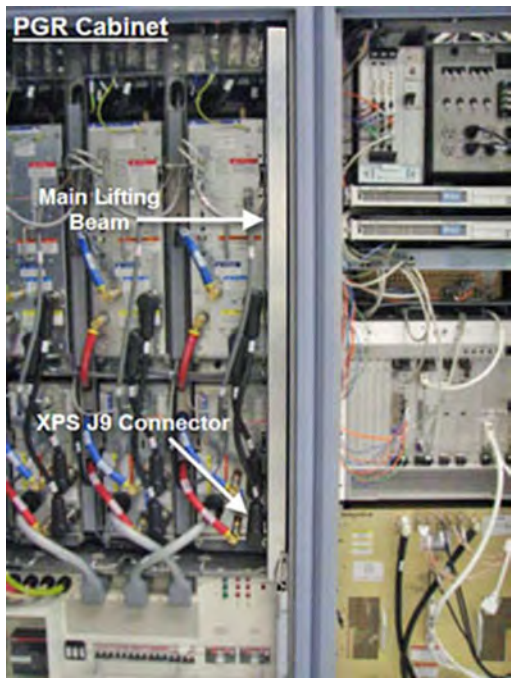
The PGR PDU/gradient subsystem must be locked and tagged out before starting this procedure. See the *MR Service Safety Manual*, PN 5452735.

NOTICE

Before disconnecting any coolant lines on the front of the XRFD unit, shut down the PPMP at the HEC.

1. Make sure the host PC is powered down.
2. Make sure the LOTO has been performed on the PGR cabinet. See the *MR Service Safety Manual*, PN 5452735.
3. Before disconnecting any coolant lines on the front of the XRFD unit, shut down the PPMP at the HEC by pressing △ and **F3** simultaneously. Then, switch the PPMP circuit breaker to **OFF**
4. Before setting up the hoist kit, detach the XPS J9 connector to properly remove the main lifting beam.

Illustration 3-23: Lifting Beam and Connector in PGR Cabinet

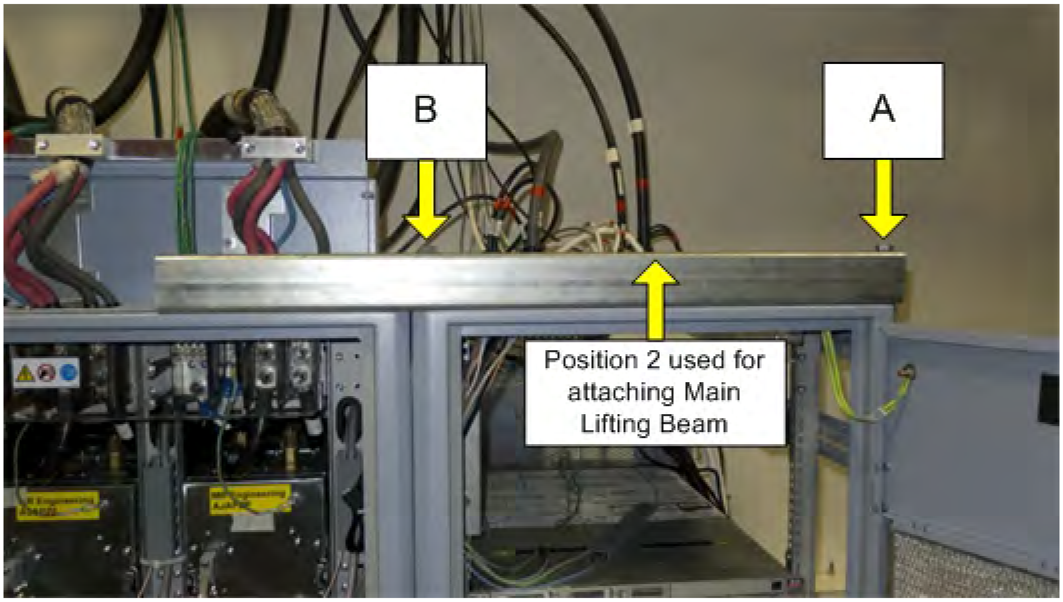


5. In the HEC, remove the support tube. The support tube is located on the right hand side of the cabinet and is secured with two mounting bolts. There are seven openings in the support tube to be used in multiple configurations.

Illustration 3-24: Support Tube - Openings Labels

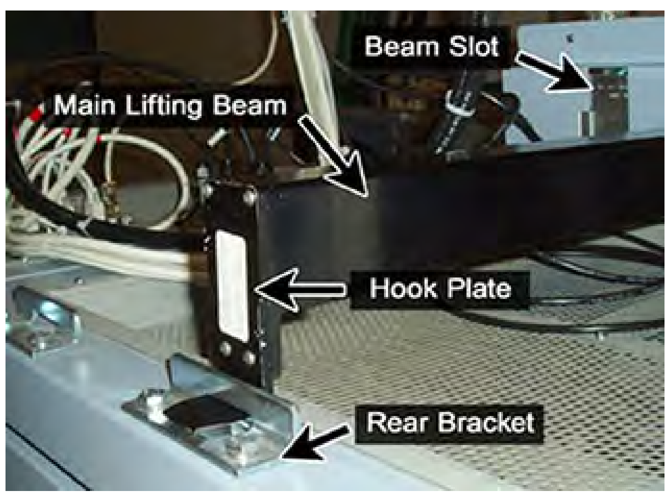
6. Place the support tube on top of cabinet. To place the support tube correctly onto the cabinet, follow the proper orientation. Secure the support tube on top of the cabinet using a wrench to tighten the two bolts onto the top.

Illustration 3-25: Support Tube in Position for RF Amplifier Removal



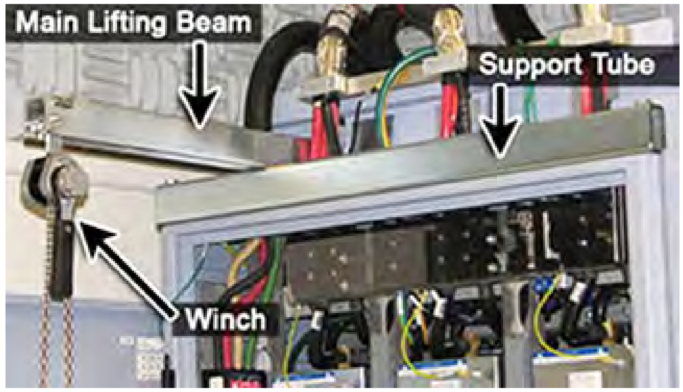
7. The main lifting beam slides onto the top of the cabinet. You may have to move cables or hoses out of the way, but none of the cables or hoses need to be disconnected.. There is a rear bracket for each position. The main lifting beam hook plate slides into the rear bracket
8. There is a rear bracket for each position. The main lifting beam hook plate slides into the rear bracket. Check to ensure that the rear bracket is tight and secure.

Illustration 3-26: Example of Cabinet Rear Bracket



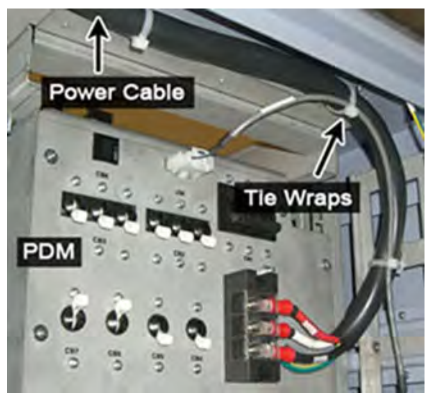
9. Slide the winch into the main lifting beam. Secure the winch in with the pin so it does not slide out. The winch is able to slide within the main lifting beam to allow proper alignment with the removed item or FRU lifting bracket.
   - Complete step 10 if PGR cabinet has a PDM.
   - Complete step 11 if PGR does not have a PDM.

Illustration 3-27: Example of Assembled Hoist Kit



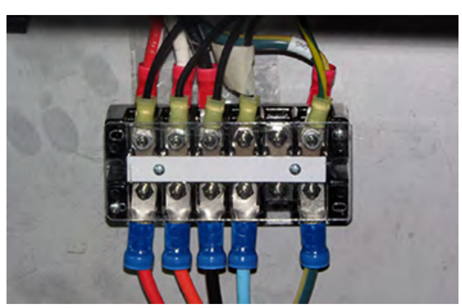
10. Power cable disconnect for cabinet with PDM.
    - Remove the tie-wraps that hold the power cable. The cable goes underneath the XRFD and along the side of the cabinet to the top of the PDM.
    - Disconnect the power cable ring terminals for the PDM terminal strip.

Illustration 3-28: Tie Wraps on PDM Power Cable



11. Power cable disconnect for cabinet without PDM after RF Amp has been removed from cabinet.
    - Roll the XRFD to the side to allow access to the cabinet base.
    - Disconnect the XRFD power cable from the terminal on inside wall of cabinet.

Illustration 3-29: RF Amp Terminal Block

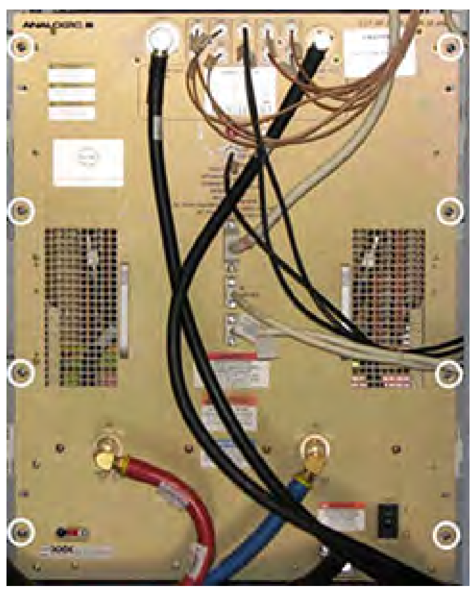


NOTE: Before continuing, be prepared to absorb any leakage from the coolant lines with a towel in hand as the line is disconnected.

Illustration 3-30: Coolant Lines Placement

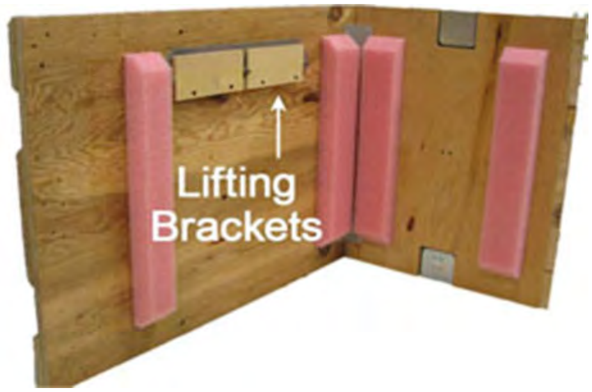
12. Disconnect the power cable at the terminal, the cables on the front of the RF Amp, and the coolant lines. Carefully move the cables and lines to the side of the cabinet, so the surrounding area is free from any obstructions when the amplifier is pulled out.

Illustration 3-31: RF Amp Attachment Screws



13. Unscrew the eight attachment screws.

Illustration 3-32: Inside FRU Crate - Lifting Brackets



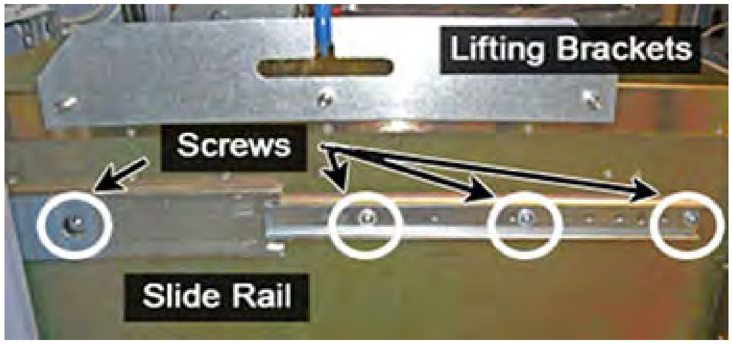
14. Using the two front handles, carefully slide out the XRFD unit (RF Amp).
15. Attach the lifting brackets (found in the new RF Amp FRU crate) to both sides of the XRFD. The attachment screws are captive screws and remain with the lifting brackets.

Illustration 3-33: Lifting Brackets with Seated Hook

16. Hook the winch and spreader bar from the hoist kit to the lifting bracket. Ratchet the winch to tighten the chain.

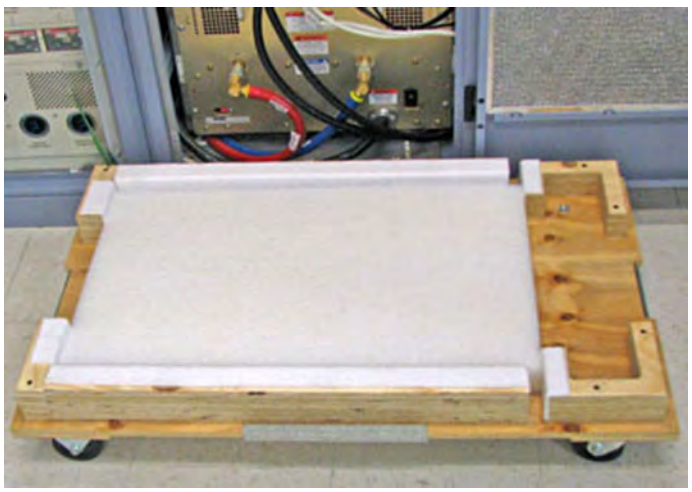
    NOTE: Make sure the hook is seated in the lifting bracket. If not, the weight can shift causing harm to the equipment or the engineer.

Illustration 3-34: XRFD Slide Rails



17. With the XRFD unit supported by the hoist kit, unscrew the four screws and remove the slide rails from both sides. The slides will be attached to the new Dual Drive RF Amp.

Illustration 3-35: Base of FRU Crate



18. The RF Amp is packaged in a wooden crate. For the XRFD, there are four parts to the FRU crate. Remove the top of the wooden crate and turn it on its base. The XRFD being removed can be placed into the top. Roll the top underneath the XRFD and lock the wheels so that the container does not move.

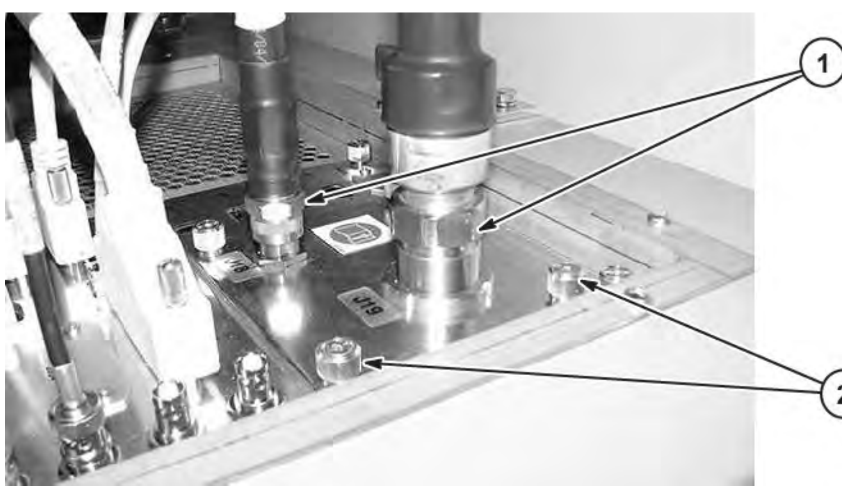
Illustration 3-36:

19. Ratchet the winch to lower the XRFD onto the platform.
20. Remove the hook from the lifting bracket.
21. Unscrew the lifting brackets from the sides of the XRFD. Roll the removed XRFD to the side.
22. It will need to be purged of water coolant before being packaged for return

5.2 PGR I/O Panel Upgrade

Refer to Cable Interconnects Upgrade for cables that are affected by this upgrade. Some cables are disconnected and reused and others are removed.

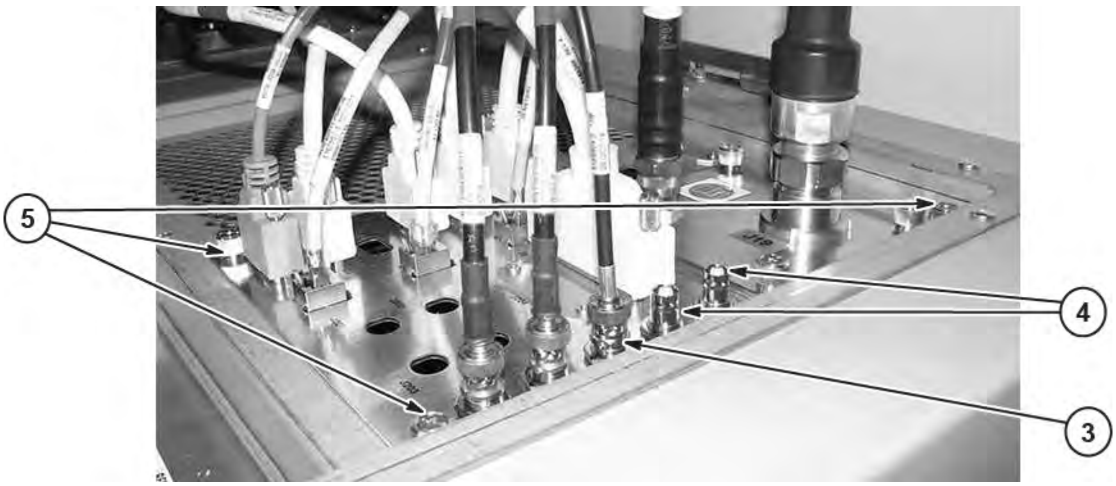
Illustration 3-37: Disconnect TX Cables from I/O Panel



1. Disconnect Runs E1001 and E1002 from J19 and J18 from the Head Body I/O Panel. Remove and discard Run E1001. E1002 will be reconnected to new I/O panel.

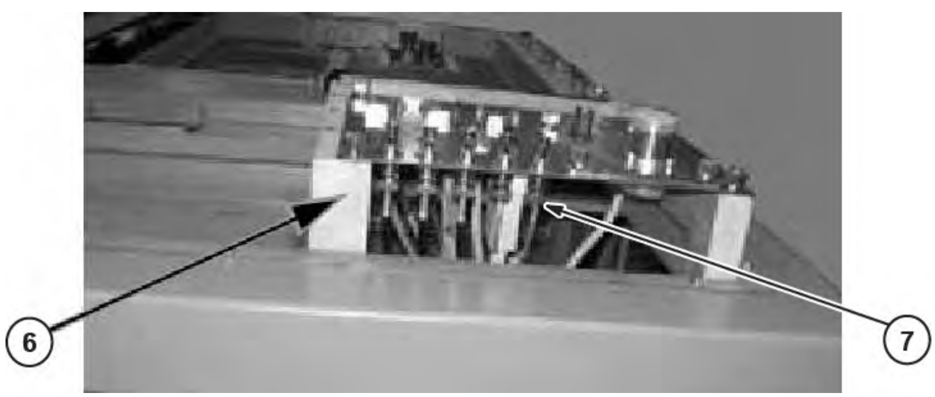
2. Unfasten the four captive screws. Lift up the panel (about 75mm) and disconnect the two cables attached to J18 and J19 from the bottom of the I/O panel. These cables will be reconnected to the new I/O panel. Discard removed I/O panel.

Illustration 3-38: Disconnect Remaining Cables from I/O Panel



3. Disconnect and discard Run E1005 from J11. A new cable will be installed to replace this one.
4. If cables are connected to J9 or J10, disconnect and discard. New cables will be installed.
5. Unfasten the four captive screws and lift I/O Panel to access cables underneath.

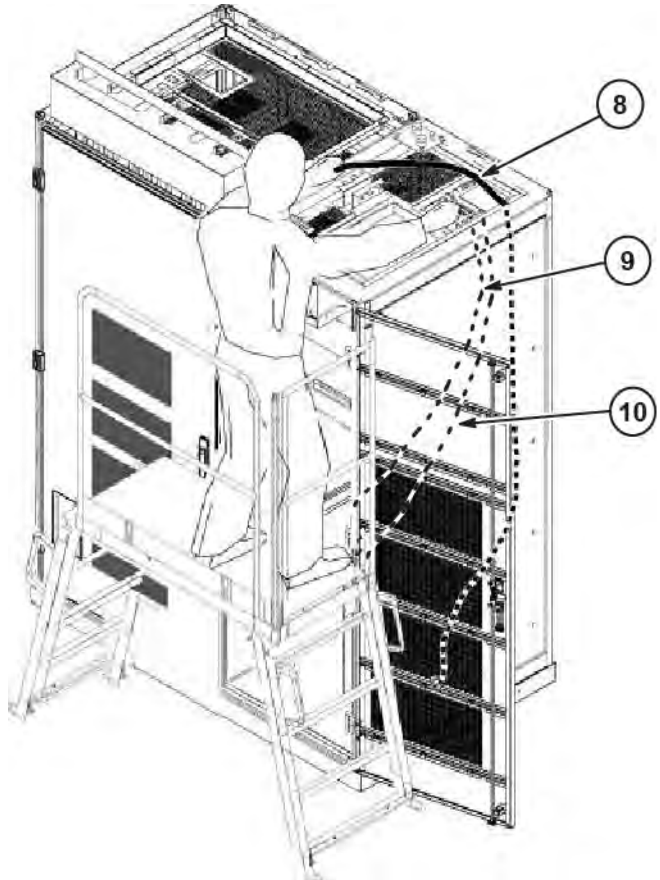
Illustration 3-39: Install Spacer Under I/O Panel



6. If possible, use some type of spacer block (example: 2in X 4in X 12in wood) to hold panel above cabinet.
7. From the bottom of the I/O panel disconnect cables attached to J8 and J11. Also, if installed, disconnect cables attached to J9 and J10.

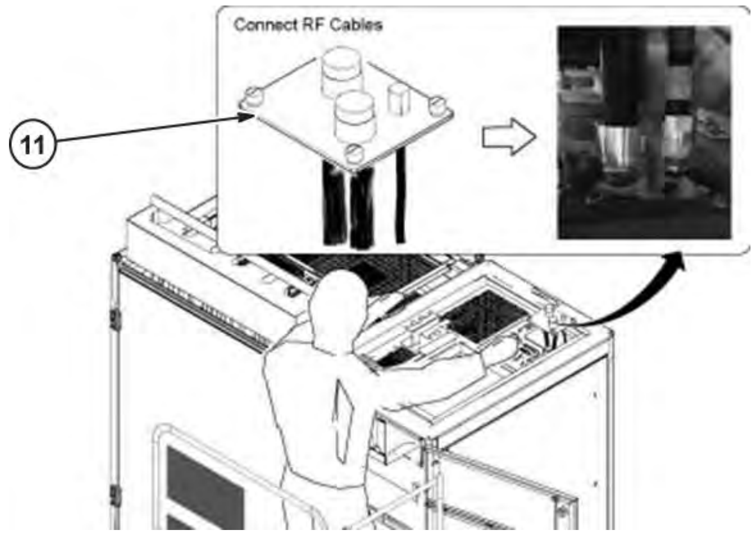
   NOTE: The cable disconnected form the bottom of the I/O panel are most likely bundled and tie-wrapped to other cables inside of cabinet and routed along side the outer wall towards the CAM and RF Amp. If they are deemed to difficult to remove, leave in place. The new cables will be routed along the same path.

Illustration 3-40: Route New Cables



8. From the I/O panel opening, route the new TX cable (5167234–5) along the same path as existing TX cables towards the location of the new RF Amp.
9. Route the four ends of the new Y-cable (5410020) along the path of existing cables to the front of the CAM Chassis. Attach 37–pin connector end to J8 on bottom side of I/O panel.
10. Route new coax cables (5419640 and 5419640–2) along the path of existing cables to the area where the top of the new RF Amp will be located. Connect as marked cable to J9 and J11 on bottom side of I/O panel.

Illustration 3-41: Install I/O panels



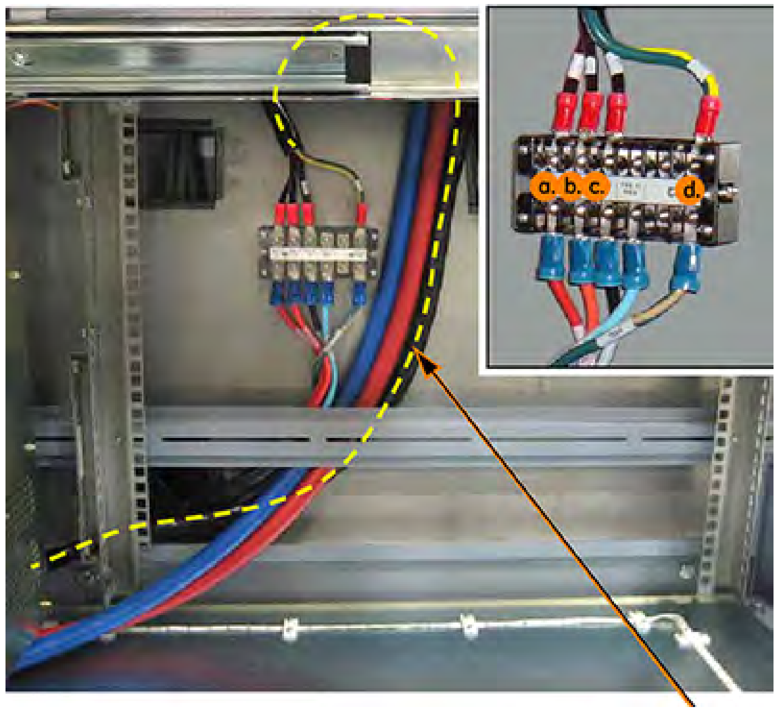
11. Place new TX I/O panel (5398351) on opening of I/O panel and connect the two existing cables to J18 and J19 and the new TX cable (5167234–5) to J23.

Illustration 3-42: Install I/O Panels and Cables

12. Fasten the I/O panel to cabinet using the four captive screws.
13. Fasten the TX I/O panel using the four captive screws.
14. Reconnect the existing TX Run E1002 to J18.
15. Route and connect the new TX Runs E1001 and E1051 to J19 and J23 as marked.
16. Route and connect the new coax cable Runs E1050 and E1005 to J9 and J11 as marked.

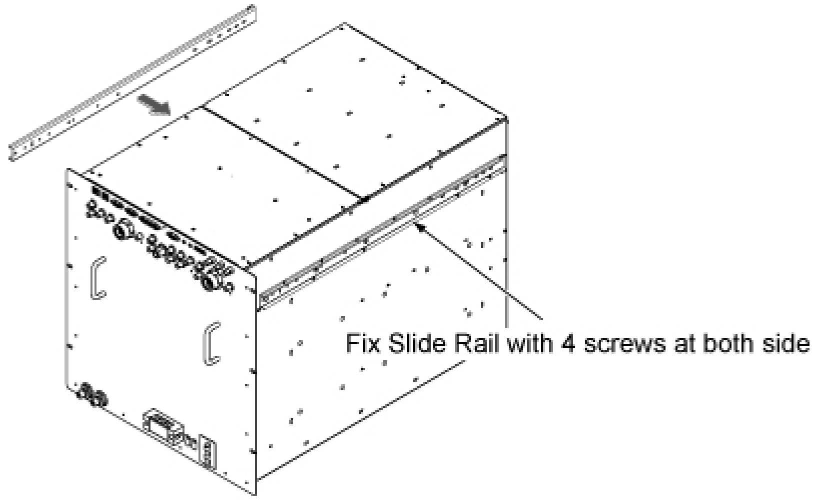
5.3 Installation of New RF Amp

Illustration 3-43: Terminal Connections



1. If PGR cabinet does not have a PDM module, then route RF Amp power cable (5399190) along inside cabinet frame and connect to terminal.
    a. Connect black wire with L1 to first terminal (a).
    b. Connect black wire with L2 to second terminal (b).
    c. Connect black wire with L3 to third terminal (c).
    d. Connect green/yellow wire to far right terminal (d).
2. Move the crate holding the new 3T Dual Drive RF Amp (5342542) to the front of the cabinet.
3. Remove the center sections of the crate.
4. Attach the slide rails to the sides of the RF Amp.

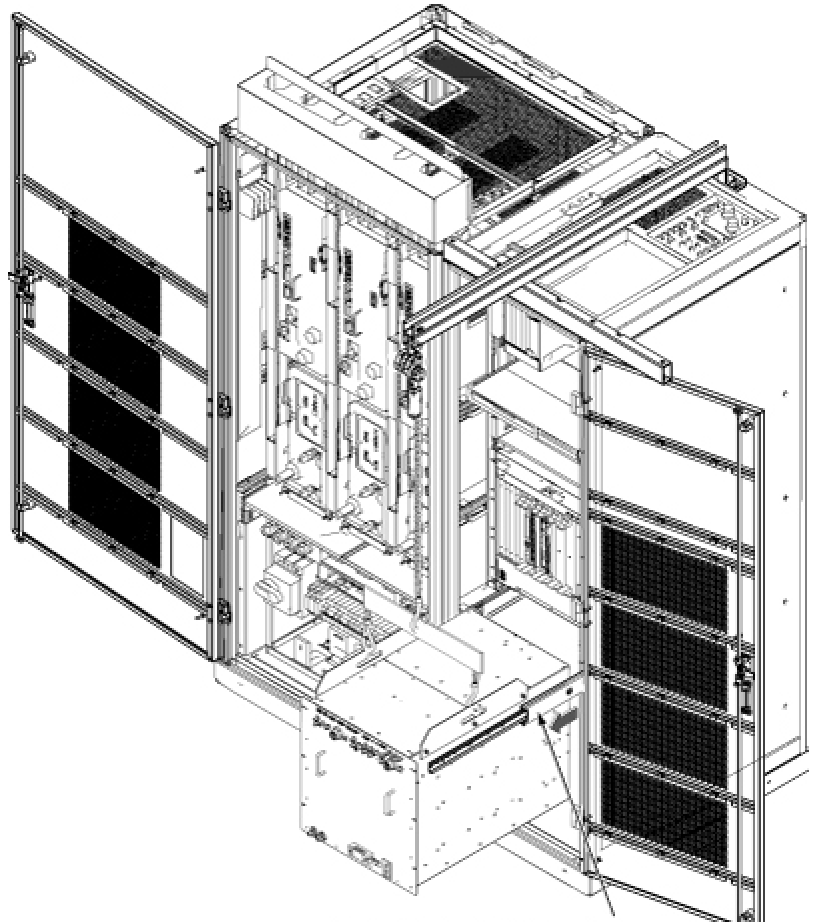
Illustration 3-44: Attach Slide Rails



5. Attach the lifting brackets to the sides of the new amplifier.
6. Attach the new dual drive RF amplifier lifting brackets to the winch and spreader bars.

   NOTE: Make sure the hook is seated in the lifting bracket. If it is not, the weight can shift, causing harm to the equipment or the engineer.
7. Ratchet the winch to tighten the chains and begin lifting the unit. Slide out the rail a short amount to determine how high to lift the dual drive RF amplifier. The dual drive RF amplifier slide rails and the cabinet slide rails must align for correct insertion.
8. When the slide rails are aligned, pull the slide rails out a short distance to get each side started. After both of the slide rails are attached, pull the slide rails onto the FRU until you hear a click. This click is a signal that the unit is docked in the rails. Ensure the dual drive RF amplifier is secure on the rails.

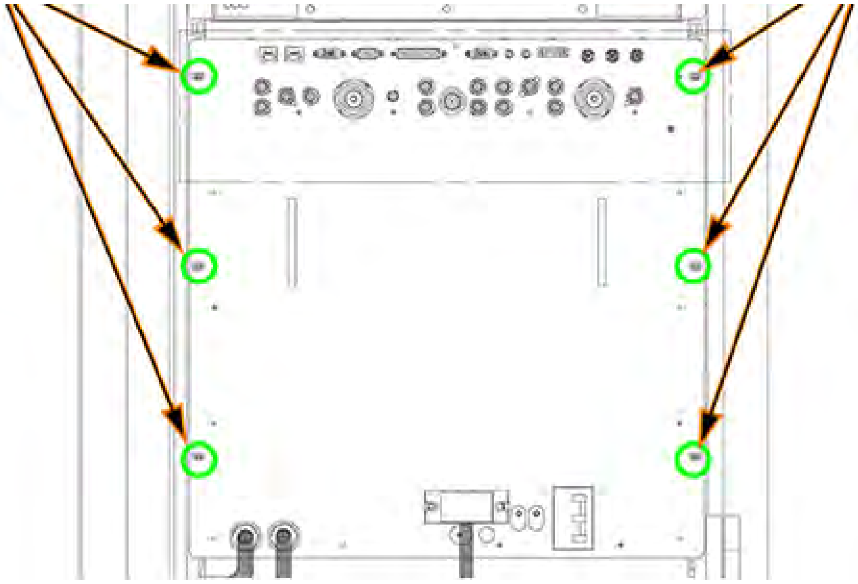
Illustration 3-45: Dock RF Amplifier in Slider Rails



9. Slowly move RF Amp into cabinet.

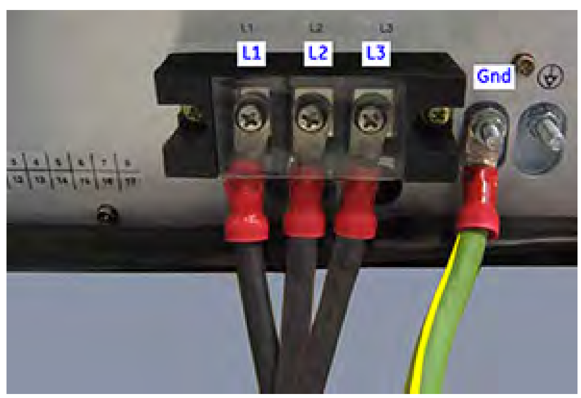
NOTE: The rails have a “soft stop” to prevent the RF Amp from moving too fast. Slowly push the RF Amp past the soft stop until it comes to a full stop.

Illustration 3-46: Securing RF Amp to Cabinet



10. Secure RF Amp to cabinet mounting rails with the six screws and washers previously removed.

Illustration 3-47: RF Amp Terminal Connections



11. Connect power cable (5399190) to RF Amp terminals L1, L2, L3 and GND as shown.

Illustration 3-48: RF Amp Cooling Lines Connections

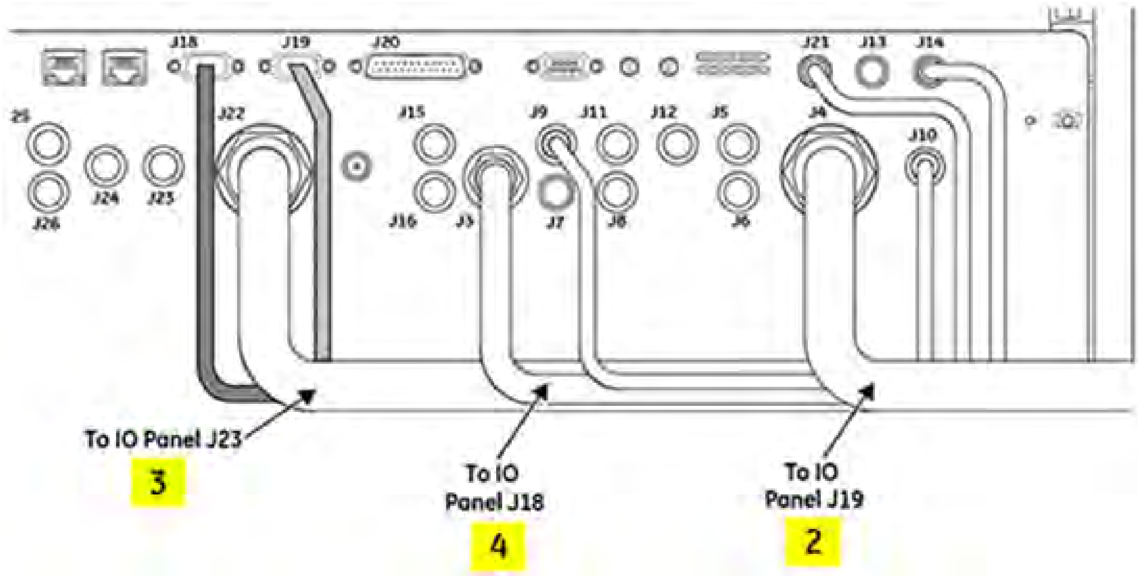
12. Connect coolant lines as follows:
    - ○ Blue (chilled) line to J1
    - ○ Red (return) line to J2.

    The following table lists the new and existing cables to be routed and reconnected to their applicable modules. The new cables are identified by an asterisk (*) after the item number.

    Table 3-1:

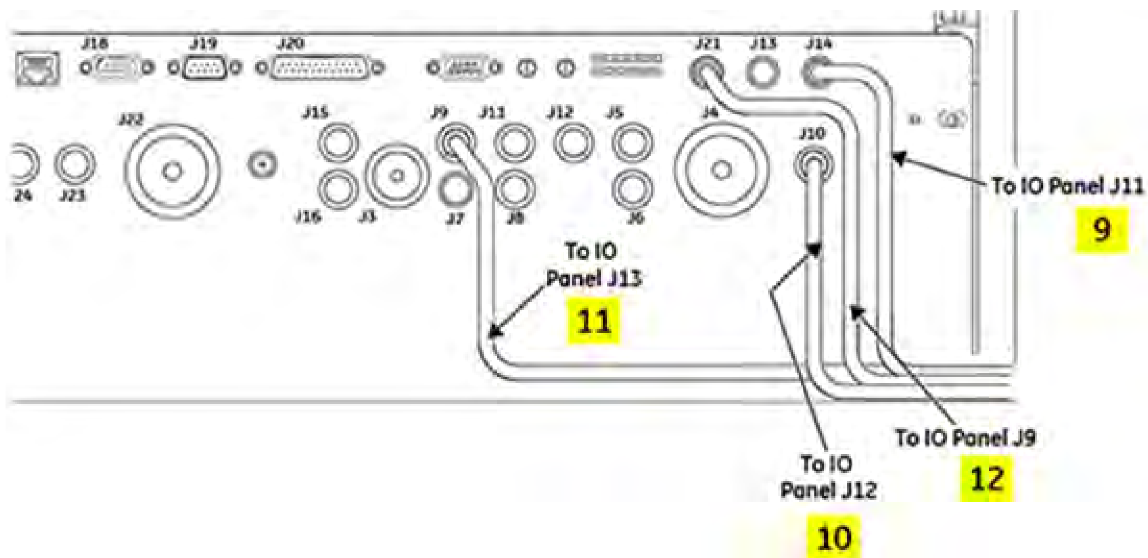
Item	Part Number	Description
2	5167234-2	Cable, Body TX, RF Amp J4 to I/O Panel J19
3*	5167234-5	Cable, Body Ch2 TX, RF Amp J22 to I/O Panel J23
4	5192690-2	Cable, Head TX, RF Amp J3 to I/O Panel J18
9*	5419640	Cable, Coax, RF IN, RF Amp J14 to I/O Panel J11
10	5176262-2	Cable, Coax, Body TR, RF Amp J10 to I/O Panel J12
11	5176262-3	Cable, Coax, Head TR, RF Amp J9 to I/O Panel J13
12*	5419640-2	Cable, Coax, RF IN Ch2 Body, RF Amp J21 to I/O Panel J9
13	5176566-2	Recon Cable, RF Amp J18 to CAM J2 (Rear of Chassis)
14	5165986	NMR Cable, CAN RF Amp J19 to I/O Panel J4
15	5166222	NMR Coax Cable, RF Amp J5 to CAM Slot 20FL J1
16	5166222-2	NMR Coax Cable, RF Amp J11 to CAM Slot 20FL J2
17	5166222-3	NMR Coax Cable, RF Amp J7 to CAM Slot 20FL J4
18	5166222-4	NMR Coax Cable, RF Amp J15 to CAM Slot 20FL J5
19	5166222-5	NMR Coax Cable, RF Amp J6 to CAM Slot 20FU J1
20	5166222-6	NMR Coax Cable, RF Amp J12 to CAM Slot 20FU J2
21	5166222-7	NMR Coax Cable, RF Amp J8 to CAM Slot 20FU J4
22	5166222-8	NMR Coax Cable, RF Amp J16 to CAM Slot 20FU J5
23*	5166222-23	NMR Coax Cable, RF Amp J23 to CAM Slot 19FL J1
24*	5166222-24	NMR Coax Cable, RF Amp J25 to CAM Slot 19FL J2
25*	5166222-25	NMR Coax Cable, RF Amp J24 to CAM Slot 19FU J1
26*	5166222-26	NMR Coax Cable, RF Amp J26 to CAM Slot 19FU J2
27	5165744	RF Amp J20 to CAM Slot 17 J8
28*	5410020	CAM Slot17/18 FL J3, FU J3, Slot 21 FU J3, FL J3 to I/O Pnl J8

Illustration 3-49: Connect TX Cables to RF Amp



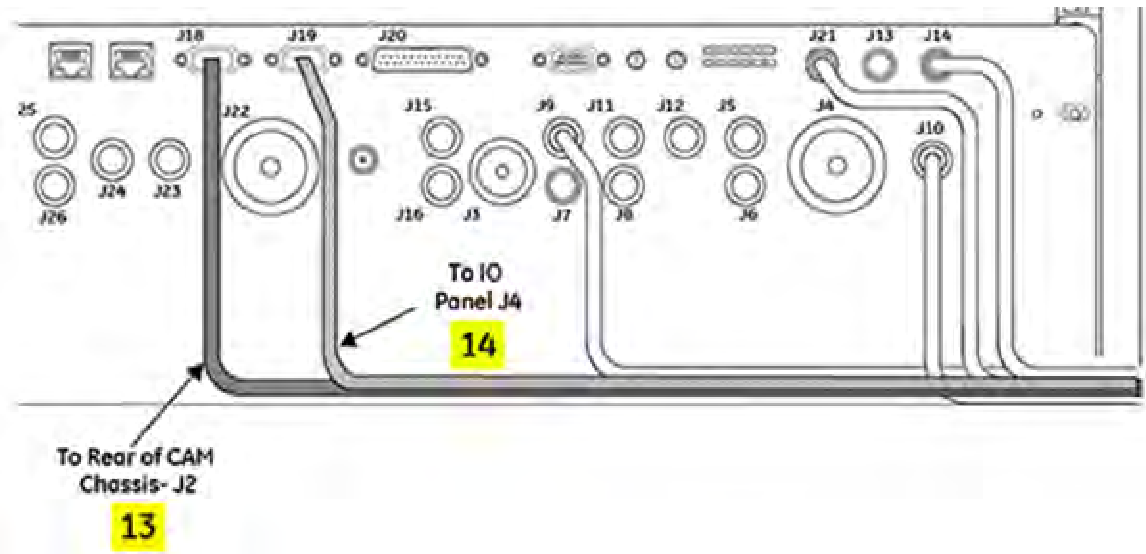
13. Connect TX cables, items 2, 3, and 4 in table, as marked. Item 3 (5167234-5) is a new cable routed from the I/O panel.

Illustration 3-50: Connect Coax Cables from I/O Panel to RF Amp



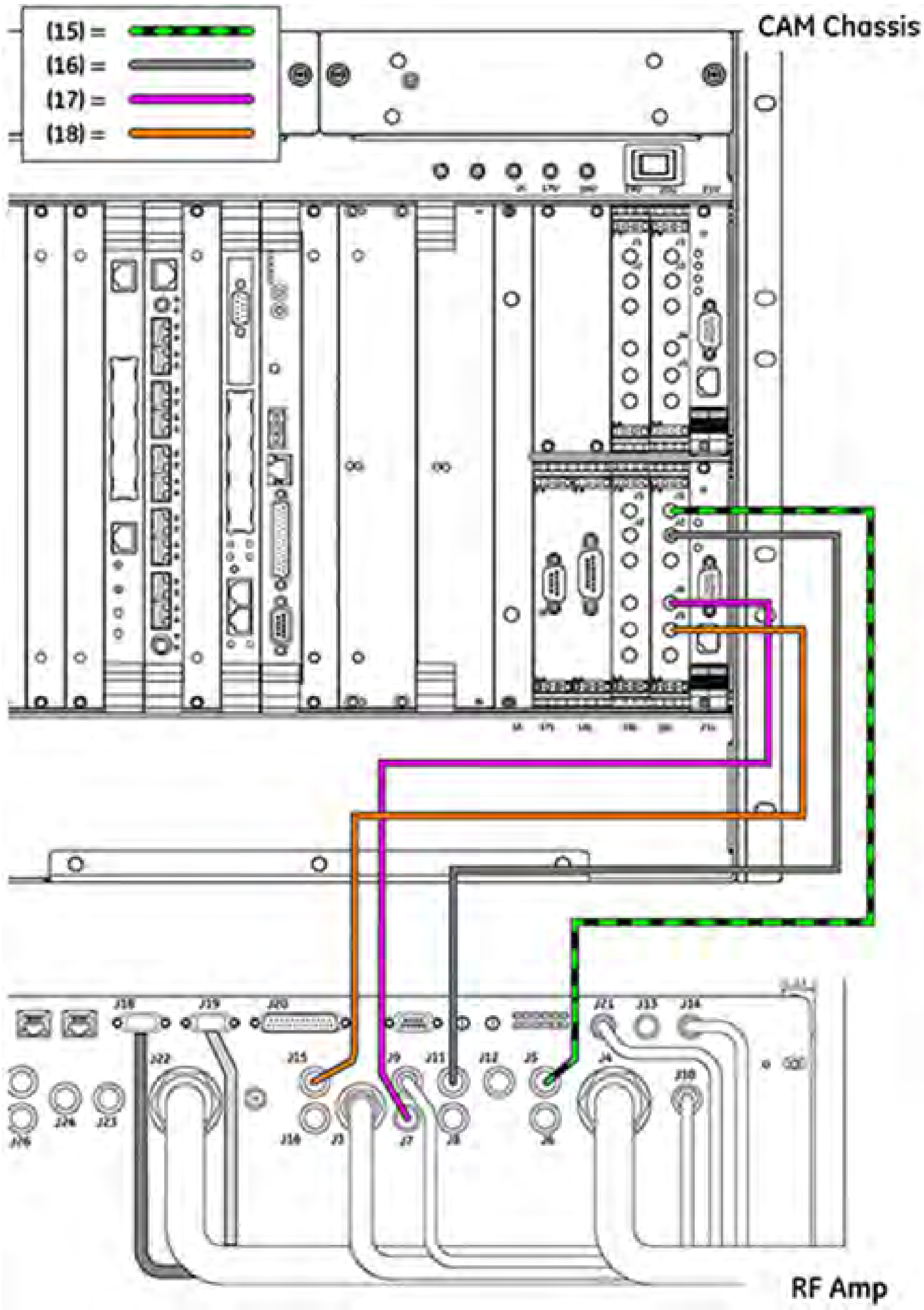
14. Connect Coax cables, items 9, 10, 11, and 12 in table, as marked. Items 9 (5419640) and 12 (5419640-2) are new cables routed from the I/O panel.

Illustration 3-51: Connect CAN cables to RF Amp



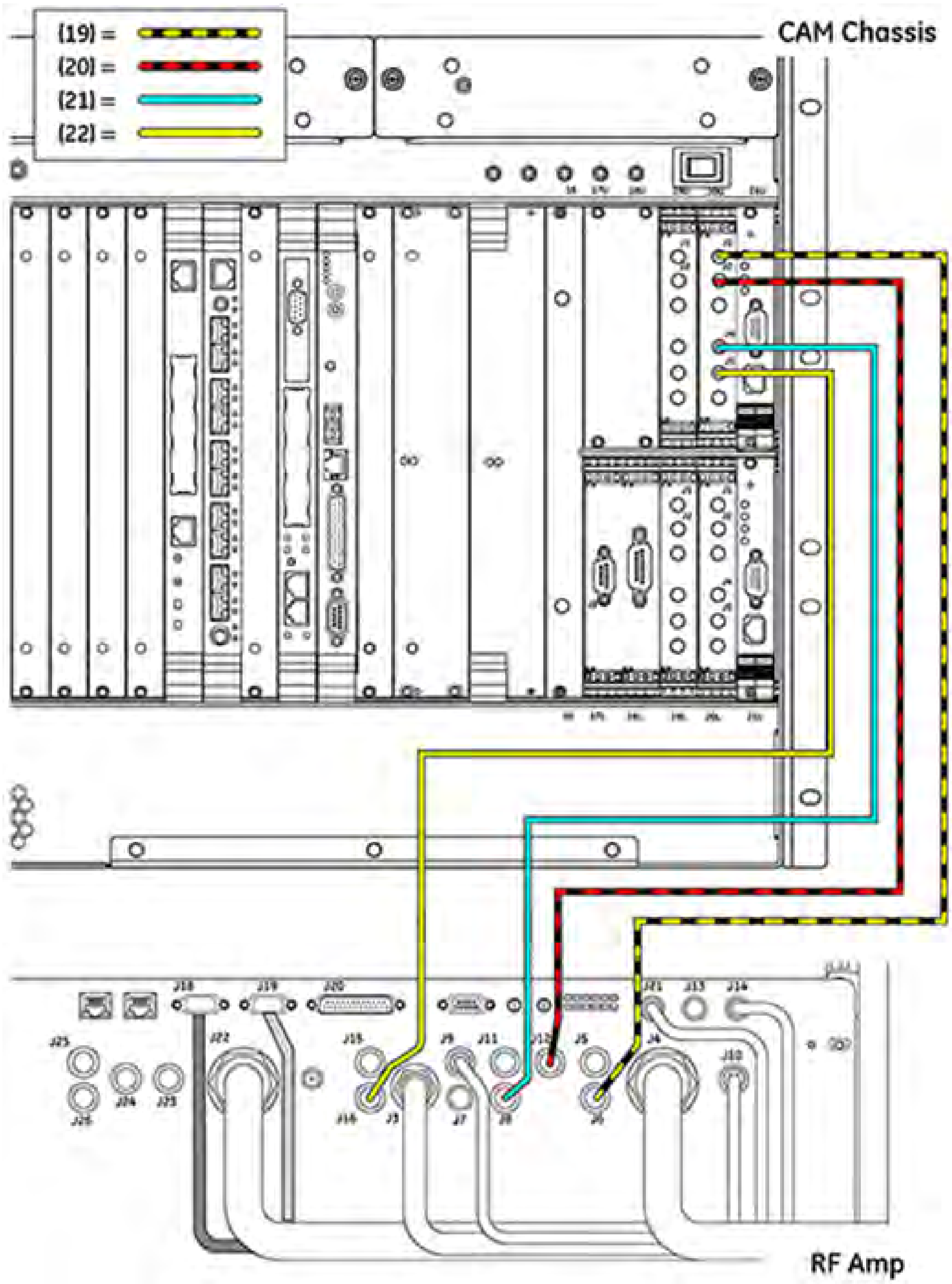
15. Connect CAN Out and In cables, items 13 and 14, as marked from to RF Amp.

Illustration 3-52: Connect Coax Cables from Slot 20FL to RF Amp



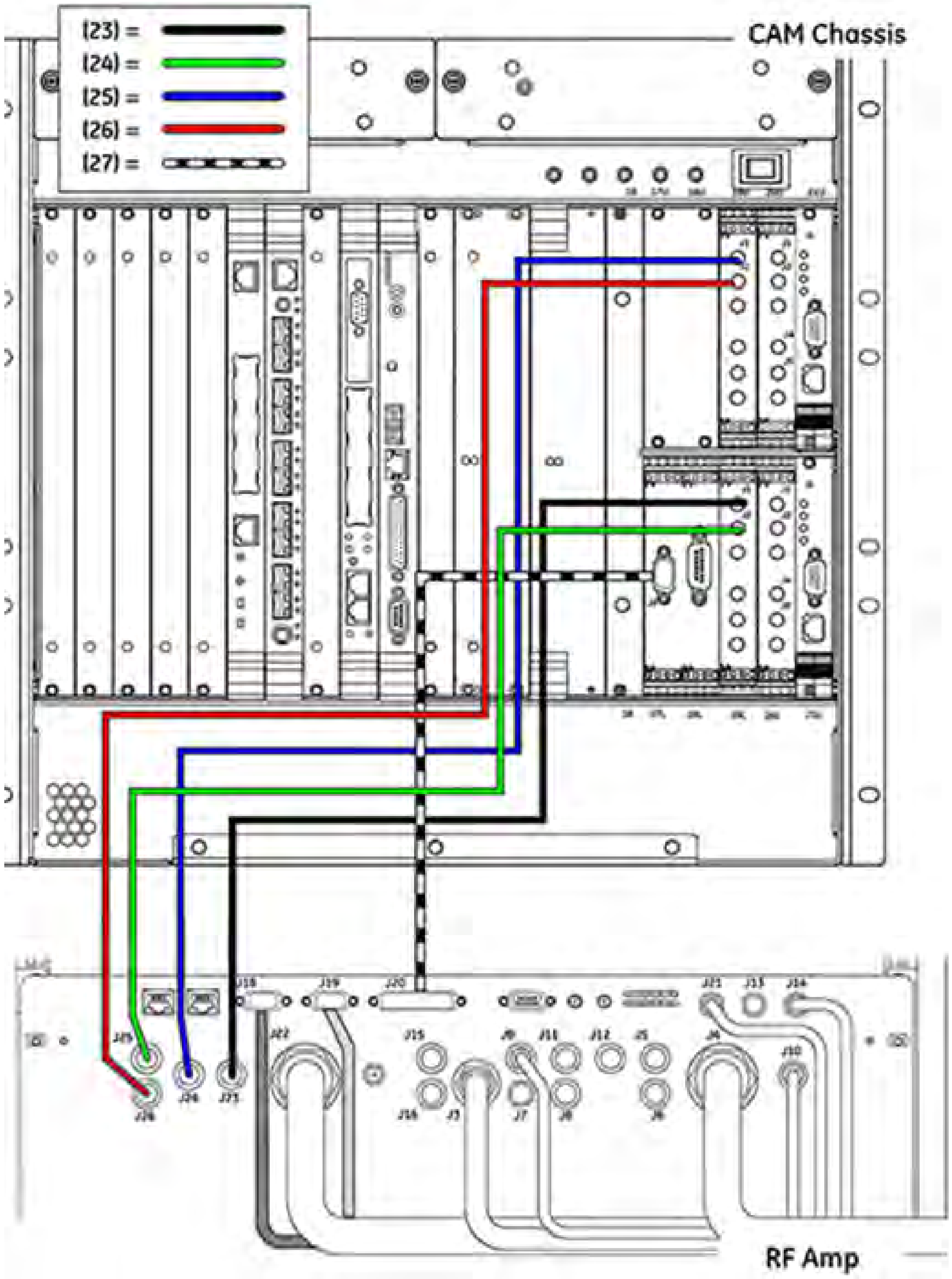
16. Route and connect NMR Coax cables, items 15, 16, 17, and 18 in table, as marked from slot 20FL to RF Amp.

Illustration 3-53: Connect Coax Cables from Slot 20FU to RF Amp



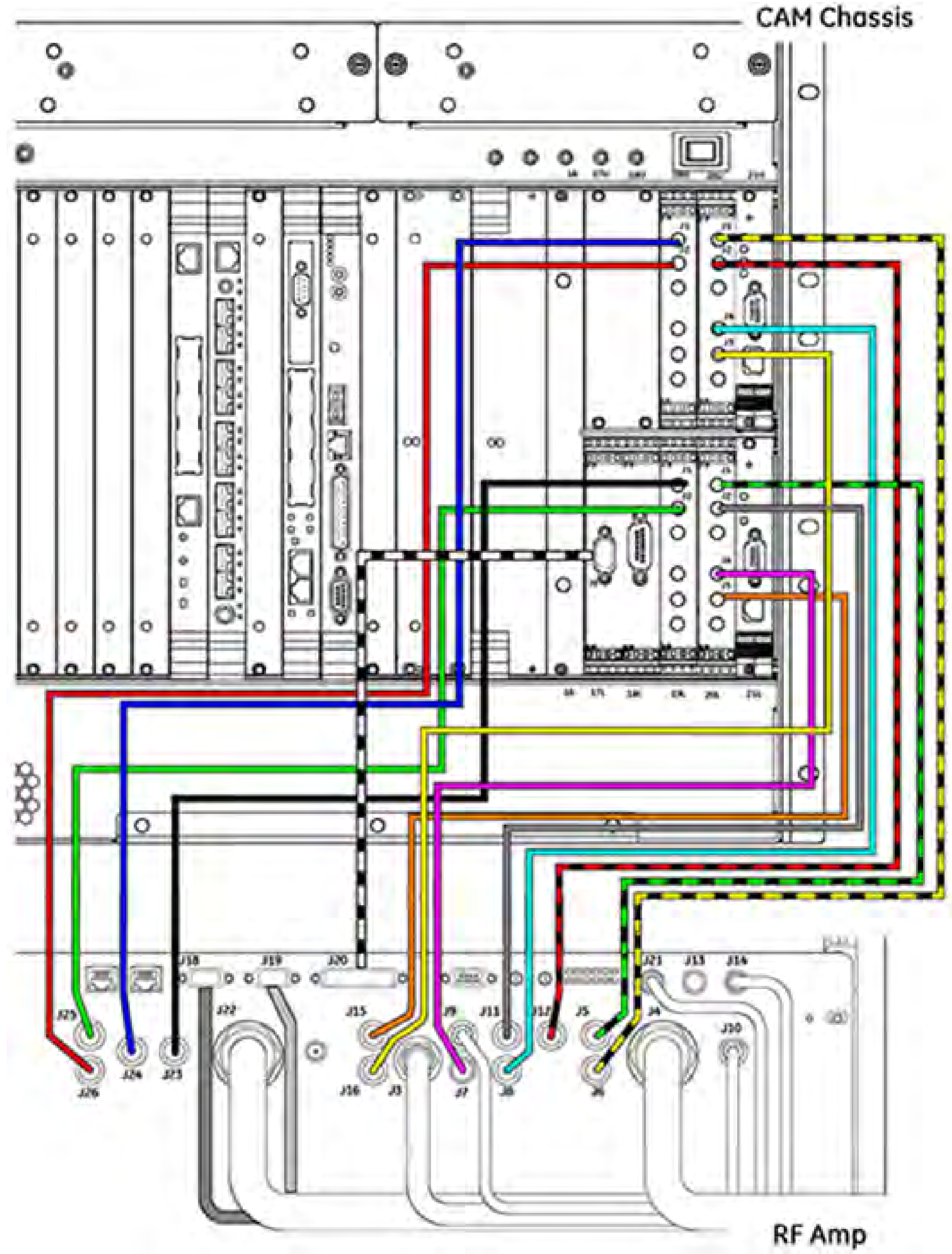
17. Route and connect NMR Coax cables, items 19, 20, 21, and 22 in table, as marked from slot 20FU to RF Amp.

Illustration 3-54: Connect Coax Cables from Slots 17 &19 to RF Amp



18. Route and connect new Coax cables, items 23, 24, 25, and 26 in table, as marked from slots 19U and 19L to RF Amp.
19. Route and connect cable, item 27 in table, as marked from slot 17 to RF Amp.

Illustration 3-55: Overview of Completed Cable Connections



20. Dress and tie-wrap cables into bundles.

5.4 Restoring Hoist Kit Components

1. Remove the winch by removing the pin from the main lifting beam and sliding out the winch. Place the winch (including hook and chains) into the hoist service kit box.
2. Remove the main lifting beam. Return the lifting beam to the PGR cabinet. Insert the hook end of the beam into the cabinet edge first, then place the bottom part of the beam in. Secure it into the cabinet with the Velcro straps and pin at the lower end of the beam.

   After the main lifting beam has been inserted in the cabinet, plug in the XPS J9 connector on the Z-axis.
3. Unscrew the support tube from the top of the cabinet. Return the support tube to the HEC and secure with the two bolts.

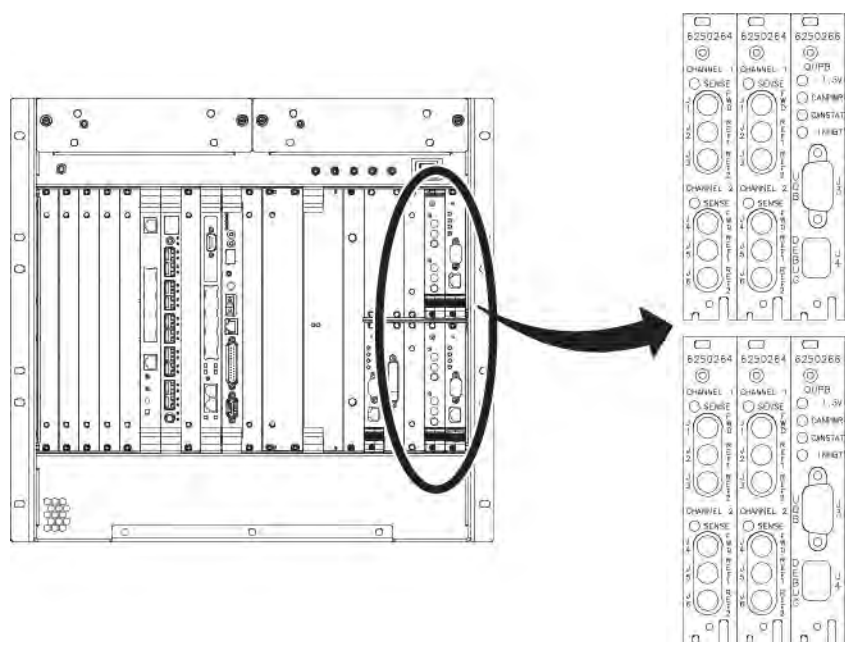
5.5 CAM Chassis Upgrade Procedures

5.5.1 MR750 Board Positions

For slots 19U, 19L, 20U and 20L,four new RF Detector boards (6250264) are to be installed.

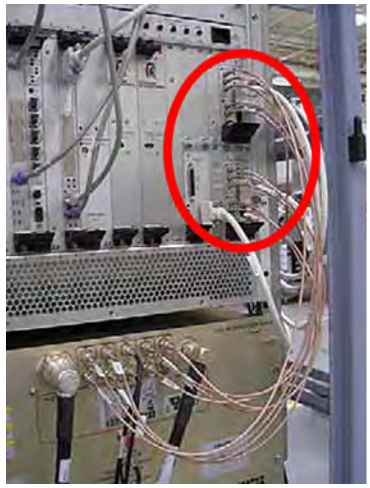
For slots 21U and 21L, two new UPM Processor boards (6250266) are to be installed.

Illustration 3-56: MR750 New Board Locations



5.5.2 Circuit Board Removal

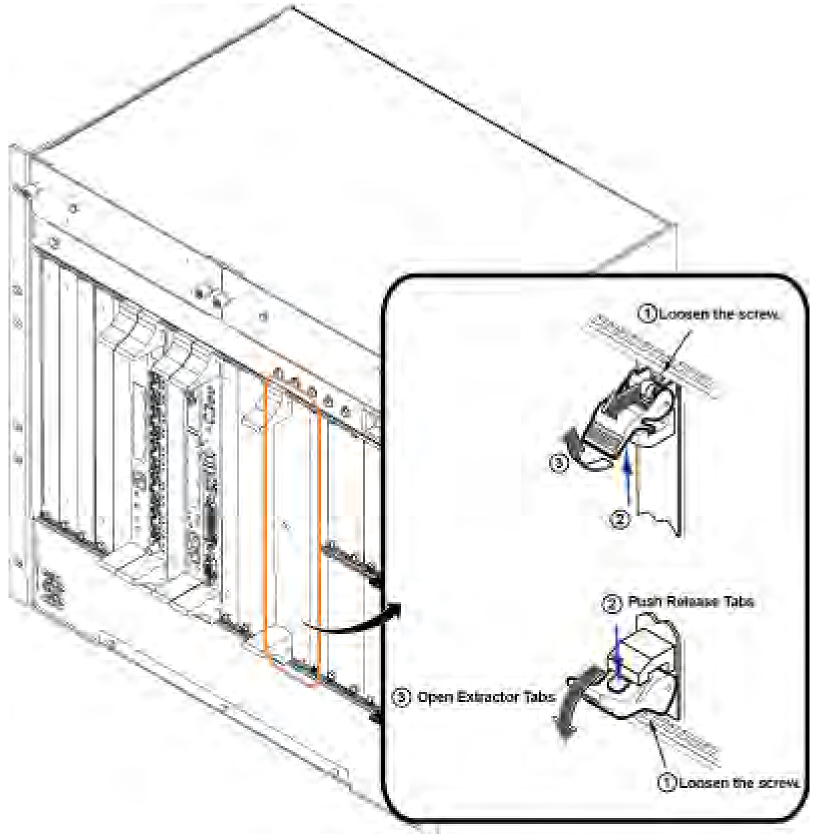
Illustration 3-57: Disconnect Cables from UPM Boards



1. Attach an ESD grounding wrist strap.
2. Where applicable, disconnect the cables (Sub D, SMB, etc.) from the faces of the boards.

   NOTE: When removing the cables, verify that the cable labels are accurate.
3. Loosen the faceplate holding screw. (These screws are captive to the faceplate, and do not need to be completely removed.)
4. Using your thumb, push in the gray release tab and gently pull the extractor tab located on the bottom of the circuit board. The lever action of the ejector disengages the circuit board connections from the chassis.

Illustration 3-58: Board Extractor Tabs



5. Being careful not to bend or bind the card in any way, slide the circuit board all the way out of the card cage.

   NOTE: Handle the circuit boards using only the faceplate, ejectors, and the edges of the board.

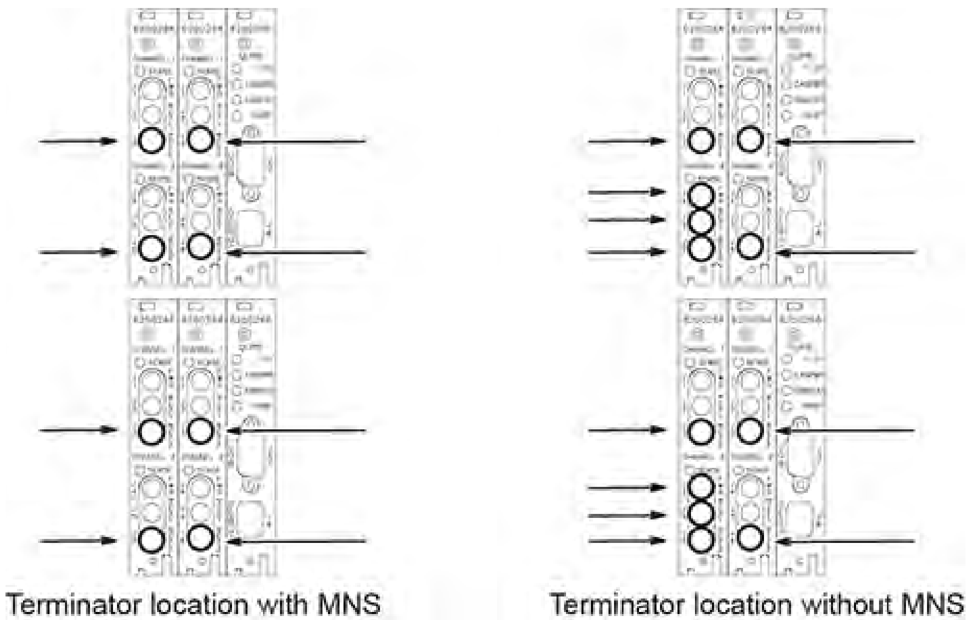
6. Place the UPM Processor boards, with the completed test procedure, in an appropriate shipping package to return for recycling. The Detector boards should be disposed of per local regulations. Refer to the *Disposed Material* section in Overview.

5.5.3 Circuit Board Replacement

1. Remove the packing material from the applicable replacement board. Position board in assigned slot and gently slide the circuit board along the slot guides into the chassis.
2. Gently engage the circuit board connectors with the backplane.
   a. If the circuit board is not properly engaging the backplane, remove the board and check the connectors for bent or broken pins.
   b. Once the board ejectors contact the CAM chassis faceplate, gently push the ejector into the center of the board. This causes the ejector to insert the board into the chassis. (Do **not** attempt to use anything other than the ejector to insert or remove the board from the chassis.)

3. Tighten, but do not overtighten, the faceplate holding screws to secure the circuit board to the card rack.
4. Reattach the cables to the face of the board.

Illustration 3-59: Terminator Locations



5. Ten 50 ohm terminators (2329754) have been supplied with the upgrade. As indicated in the illustration:
   a. For systems with MNS, only eight (8) terminators are required and installed as shown.
   b. For systems without MNS, twelve (12) terminators will be required — the ten that were delivered with the upgrade and the re-use of two terminators that were already installed prior to the upgrade.

5.5.4 Return Removed Modules

All discarded material (i.e. removed/discarded hardware, packing material) must be returned in accordance with GE recycling policy. Refer to Discarded Material Return Policy

Chapter 4 Power On Procedures

1 Removing LOTO

After completing the mechanical upgrade procedures, complete the following steps for powering up the system.

1. Prepare for reenergizing the equipment.
2. Remove personal LOTO device(s) from the MDP
3. Switch on the MDP's PDU breaker.
4. Remove personal LOTO device(s) from the front panel of the PGR PDU.
5. Rotate the larger main breaker clockwise to the *RED I (ON)* position.
6. Press the **EMO RESET** switch.

2 Restarting Host PC

Restarting the Host PC shuts down the system safely, and automatically starts the system up again. Restarts are often required to finalize changes or updates.

To restart the system, click the arrow on the *Tools High Level Access* tab, select System Restart, and click [Yes] to confirm the restart.

Chapter 5 Final Setup Procedures

1 Checks and Calibrations

1.1 Checks and Calibrations Flowchart

After the hardware has been installed, proceed with the following to complete the Dual Drive upgrade.

1. Software Load and Reconfiguration. Refer to Section 1.2
   - Software Load (if needed)
   - Configure for Dual Drive
   - Check Guided Install
2. TR DD Calibration. Refer to Section 1.3
3. RF Amp/UPM calibration and UPM functional check for head and body. Refer to Section 1.3
4. RF Body Coil Centering Pad Installation. Refer to: Section 1.4
5. Body Coil and Endbell Gap Check. Refer to Section 1.5
6. Body Coil Centering. Refer to Section 1.5
7. Cable Swap Test. Refer to Section 1.6
8. Run DD Quad Cal. Refer to Section 1.7
9. Body Coil Tuning. Refer to Section 1.8
10. Remaining Functional Checks and Calibrations. Refer to Section 1.9
11. Perform SaveINFO. Refer to Section 1.10

1.2 Software Load and Reconfiguration

If needed, load the latest software to enable the Dual Drive feature. Software revision DV24 and prior releases will not support Dual Drive.

Perform one of the following:

- After loading the latest software, go into “Guided Install” and then change the RF Amp type from “Single Drive” to “Dual Drive”. Change the UPM type to QUPM type under Hardware Reconfiguration. Then perform a **RestoreInfo**. The ICW will come up in “Upgrade Mode” to guide you through the calibrations that are needed for the Dual Drive Upgrade.
- If the latest software that supports Dual Drive is already loaded, go into “Guided Install” and then change the RF Amp type from “Single Drive” to “Dual Drive”. Change the UPM type to QUPM type under Hardware Reconfiguration. Then perform a **RestoreInfo**. The ICW will then come up in “Upgrade Mode” to guide you through the calibrations that are needed for the Dual Drive Upgrade.

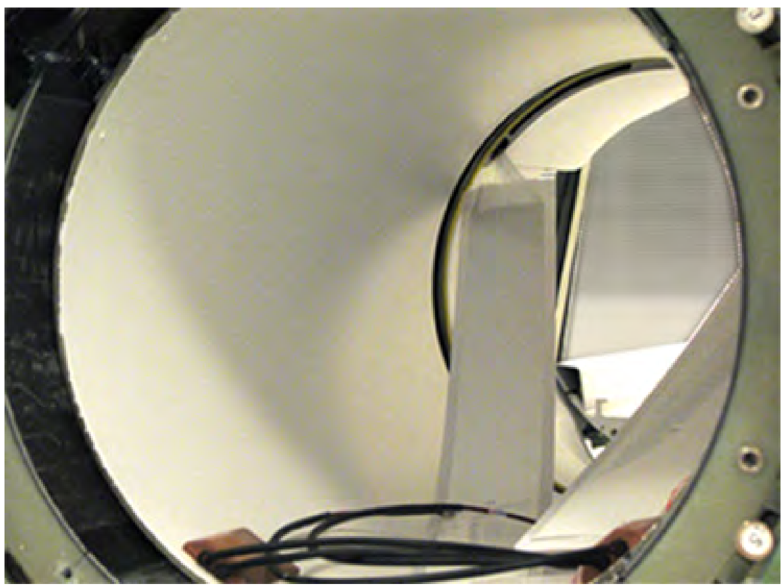
1.3 TRDD Calibration and RF/UPM Calibration and Functional Checks

Perform TRDD Calibration and RF/UPM Calibration and Functional Checks per the ICW and latest Service documentation.

1.4 RF Body Coil Centering Pad Installation

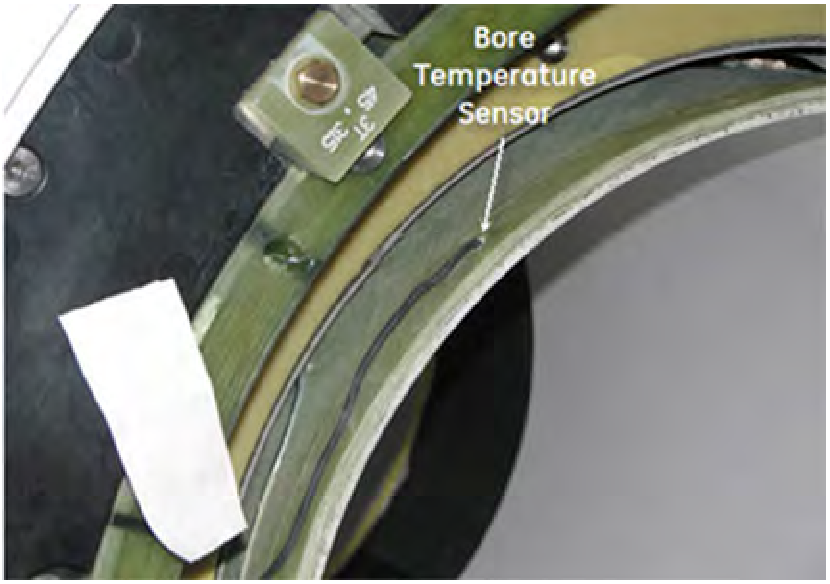
1. Remove the Front End Bell per the **Front End Bell Removal and Installation** service procedure included in the MR750 Service Methods DVD under **Replacements / Magnet Enclosure**.
2. Disconnect the two DD bias cables in the front, patient end.
3. Disconnect the two I and Q drive cables from the rear of the RF body coil.
4. If installed, disconnect bore lights from the RF body coil. There are four nylon push rivets that must be popped out before removal.

Illustration 5-1: Bore Lights Removed



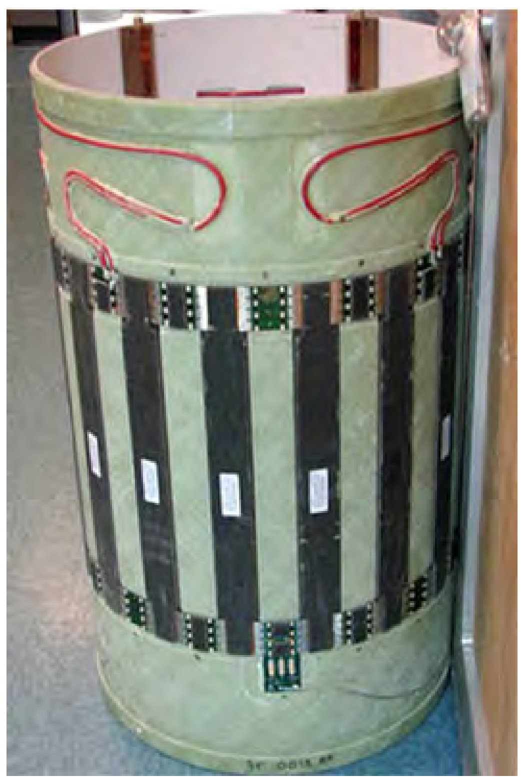
5. Partially slide out the RF body coil, and disconnect the bore temperature sensor.

Illustration 5-2: Bore Temperature Sensor Cable



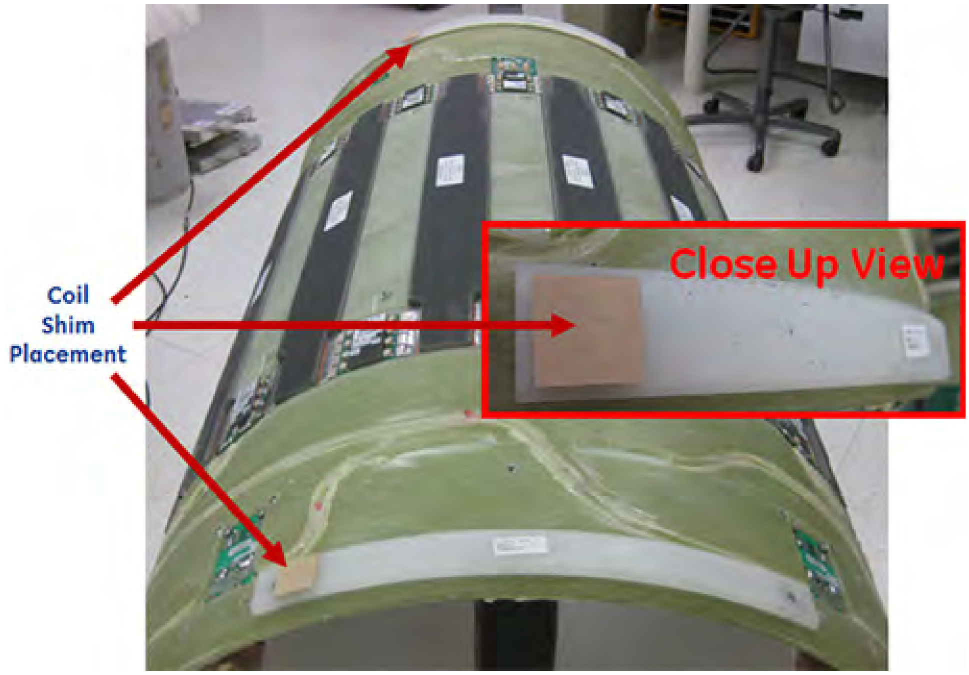
6. Fully remove the RF body coil and set it on the patient end.

Illustration 5-3: RF Body Coil on Patient End



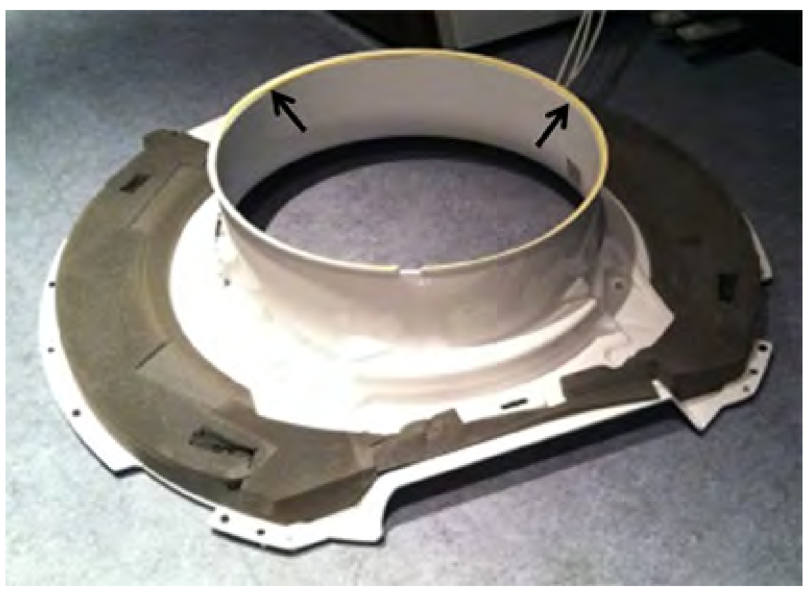
7. Visually inspect the gradient coil and RF body coil to confirm whether additional shims have been previously installed between the two coils to help with RF body coil centering. If shims have been installed, remove and discard them. The shims might be stuck to the body coil centering spacers, or they might be stuck to the gradient coil acoustic liner

Illustration 5-4: Possible Shim Locations on RF Body Coil Centering Spacers



8. If the endbells have their original yellow foam installed, it will need to be replaced per FRU kit 5498152. If replacement is necessary, remove the Rear End Bell per the **Rear End Bell Removal and Installation** service procedure included in the MR750 Service Methods DVD.

Illustration 5-5: Original Yellow Endbell Foam



9. Remove both RF Body Coil Centering Spacers, 5177757.

Illustration 5-6: RF Body Coil Centering Spacer — Service End

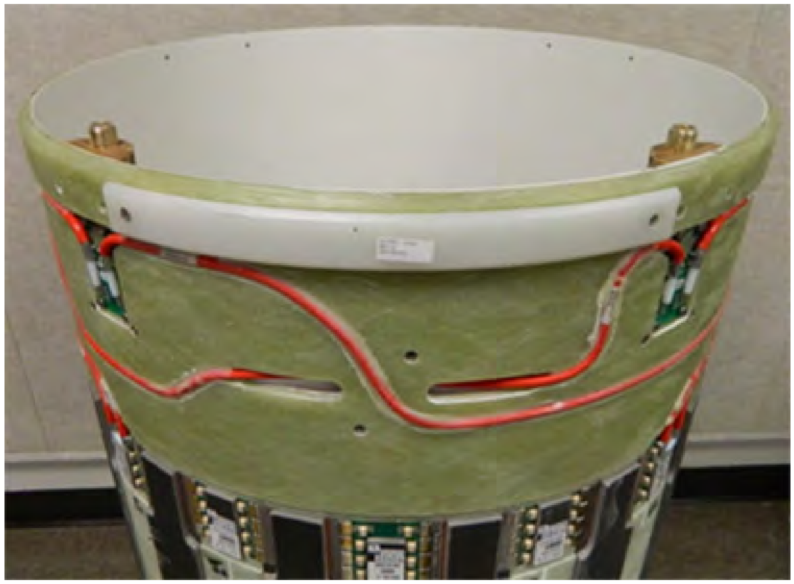
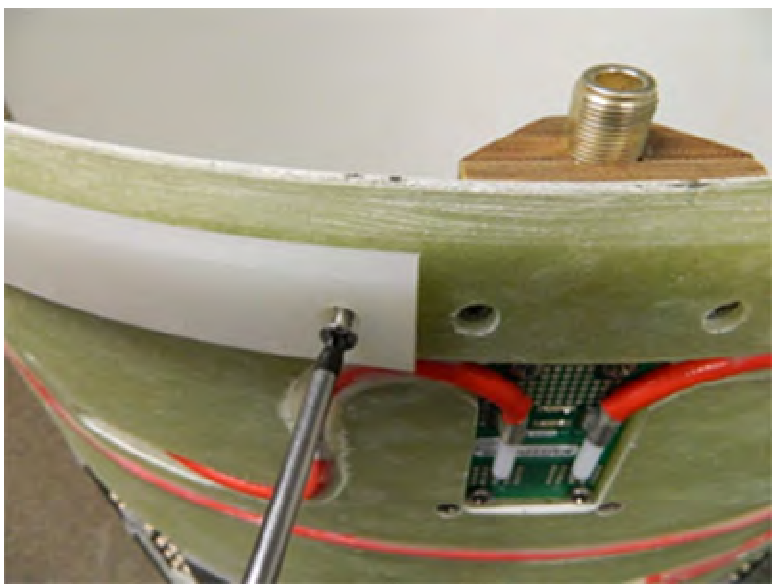


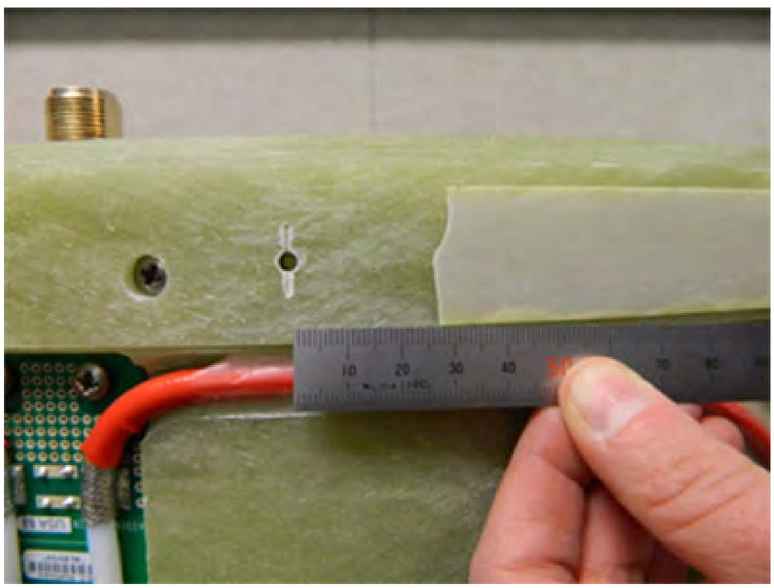
Illustration 5-7: Removing the RF Body Coil Centering Spacer



10. 2.10 If the original centering spacer adhesive sticks to the RF Body Coil, remove a minimum of 25mm beyond each mounting hole using the Body Coil Lift Tool, 5502123. Refer to the next two illustrations.

Illustration 5-8: Removing the Centering Spacer Adhesive

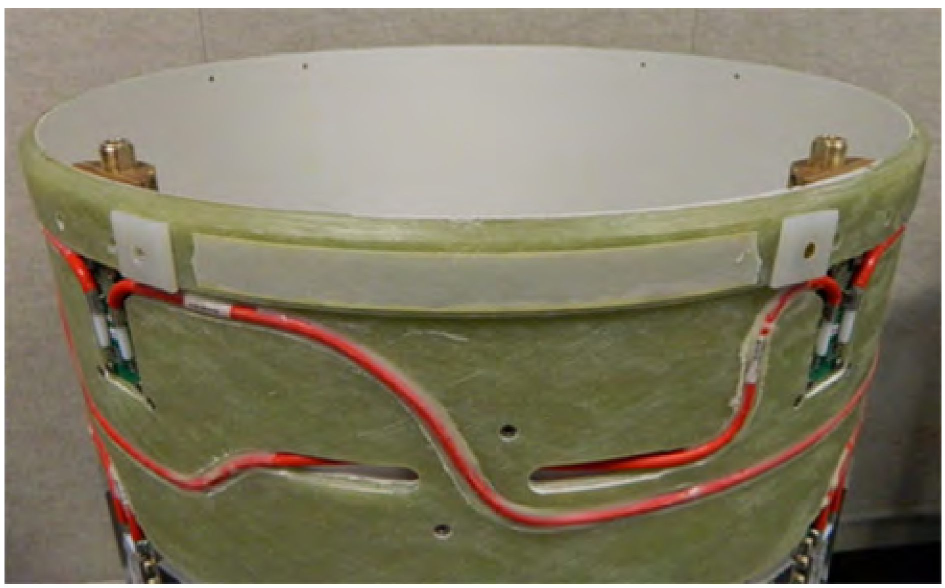
Illustration 5-9: Centering Spacer Adhesive Removed Minimum of 25mm Beyond Mounting Hole



11. Install four centering pads, 5494359, at each of the original centering spacer attachment points using M4x12 Phillips Flat Head Screws. Installing the noted centering pads will position the body coil in a logical starting point for performing the coil centering procedure.

Illustration 5-10: Centering Pads Installed (View with no Residual Adhesive)

Illustration 5-11: Centering Pads Installed (View with Residual Adhesive)



12. If previously removed, re-install the rear endbell per the **Rear End Bell Removal and Installation** service procedure included in the MR750 Service Methods DVD.
13. Re-install the RF body coil back into the magnet bore. Before sliding the RF body coil in all the way, reconnect the bore temperature sensor to the coil, referring to **Room Ambient and Bore Temperature Sensor Replacement.** This procedure is located on the latest MR750 Service Methods DVD under **Replacements / Magnet Enclosure**.
14. Check the RF body coil levelness (with respect to the x-axis) using the tubular level, 5450084 which is included in the leveling kit. The bubble is to be within the two black lines when placed on the small RF cable cover between the two drive cable bulkheads in the rear of the body coil.

Illustration 5-12: Placement of Tubular Level

15. Press the RF body coil rearward so there is a 2.0 to 2.5 mm gap between the coil and the rear end bell. The ideal gap is 2.0 mm.
16. Re-install the front endbell per the **Front End Bell Removal and Installation** service procedure included in the MR750 Service Methods DVD.
17. Reconnect the DD bias lines and the two I and Q drive cables.
18. If the endbell to body coil gaps were previously in spec, re-check them to confirm they are still in spec (2mm to 2.5mm), or note if they are now out of spec.

NOTE: When adjusting the coil to achieve levelness, make sure to have two people rotate the coil from the front and back so as not to “corkscrew” the coil (i.e. the back rotates but the front does not rotate). A tip to rotate: If one pushes up on the coil with one hand, the pressure/friction from gravity is less between the body coil and gradient coil, and it is easier to make slight rotation adjustments.

NOTE: Remember to check for body coil levelness with the tubular level every time you move the body coil or endbells. Re-adjust for levelness as needed.

1.5 Body Coil Endbell Gap Check and Body Coil Centering

Perform Body Coil Endbell Gap check and Body Coil Centering using the **RF Body Coil Replacement Procedure**. This procedure is located on the latest MR750 Service Methods DVD under **Replacements / Coils**.

1.6 Cable Swap Test

The Cable Swap Check is performed to determine if a phase or magnitude error is being caused by the RF Cabling or the Body Coil. Dual Drive Quadrature data is taken with cables swapped. The change in values will then determine if the error is within the RF cabling.

1. Perform **Dual Drive Quadrature Tool** located on the MR750 Service Methods DVD, under Adjustments/Calibrations/System. Run ONLY the **Test Mode** at this time Record the Phase and Magnitude results.

Phase ____________

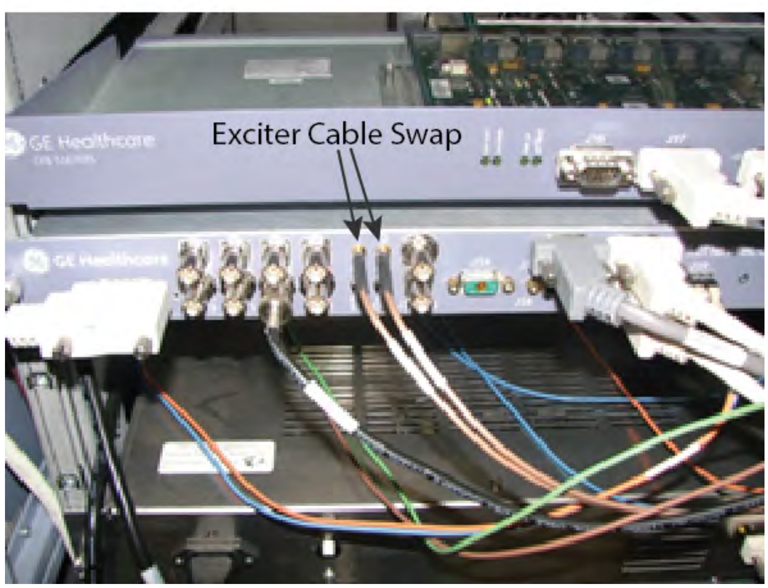
Magnitude ____________

2. Table movement can be emulated out in order to scan without a functioning LPCA. If needed, emulate the table by doing the following:
    a. Open a terminal window
    b. Type **cd /w/config**
    c. Type **gedit mgd_stage**
    d. Type **T** (using upper case) in order to emulate the table
    e. Perform TPS Reset
    f. Open terminal window and type **cd /usr/g/tools/dd_quad_tool**
    g. Type **./disablePosCheck**. (this step is needed to disable the Centering check that is part of the Dual Drive Quadrature check
3. Perform Dual Drive Quadrature Tool in [TEST] mode
4. Record Phase and Magnitude results

    Phase Error____________

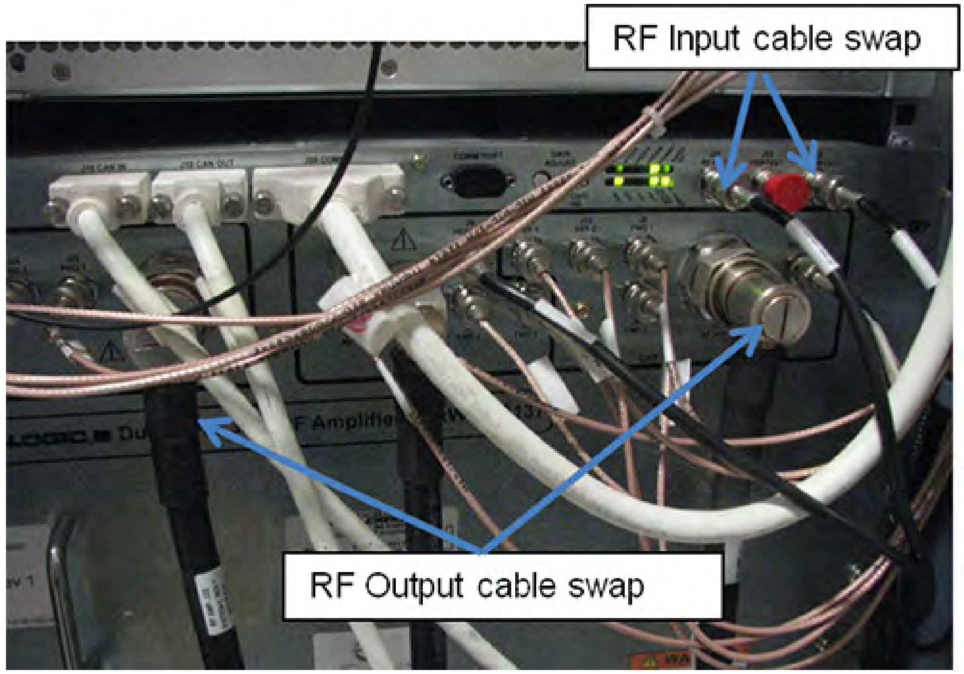
    Magnitude Error____________
5. Set the RF Enable switch on the exciter (located in the top of the PEN cabinet) to Disable (down).
6. Swap the cables at the Exciter, I and Q output. See illustration below.

Illustration 5-13: Cable Swap at Exciter



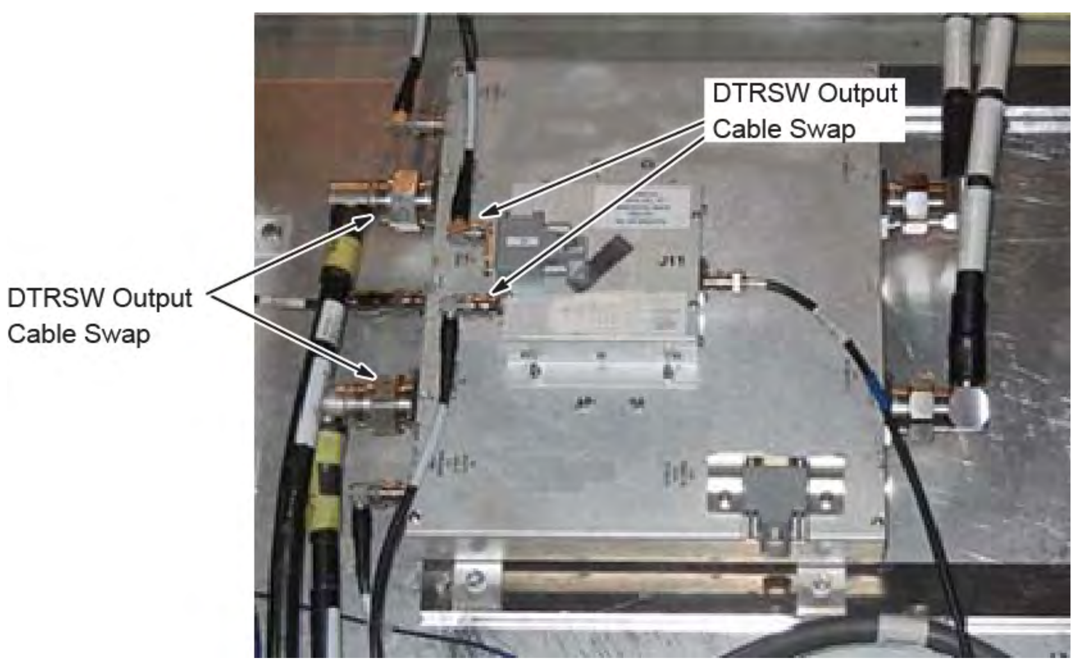
7. Swap the cables J14 and J21 at the input of the RF Amplifier

Illustration 5-14: Cable Swap at RF Amp



8. Swap the cables J4 and J22 at the output of the RF Amplifier
9. Swap the RF Input cables at the DTRSW, J1 and J4. See above Illustration.
10. Set the RF Enable switch on the exciter (located in the top of the PEN cabinet) to ENABLE (up).

Illustration 5-15: Cable Swap at DTRSW



11. Perform **Dual Drive Quadrature Tool** located on the MR750 Service Methods under Adjustments/Calibrations / System. Run the Test Mode only at this time
12. Record the Phase and Magnitude results.

    Phase Error____________

    Magnitude Error____________
13. Compare the phase and magnitude results taken in step 1 to those in step 14. If phase difference is less than 7 degrees, and magnitude difference is less than 5 counts, the cable swap test is complete.. If the difference is greater troubleshoot using **Quad Calibration Troubleshooting**, which is located on the Service Methods.
14. Set the RF Enable switch on the exciter (located in the top of the PEN cabinet) to DISABLE (down).
15. Return all cables to original position when complete.
16. Set the RF Enable switch on the exciter (located in the top of the PEN cabinet) to Enable (up).

1.7 Dual Drive Quadrature Test

NOTICE

When running Dual Drive Quadrature Test, all endbells MUST be on and fully secured to the magnet to ensure that the RF Body Coil is in the proper position.

NOTICE

Make sure that the BRIDGE IS NOT TOUCHING the Body Coil.

Perform **Dual Drive Quadrature Tool** located on the MR750 Service Methods under Adjustments/Calibrations / System. Run the Test Mode only at this time Record the Magnitude value, as this will be needed for Body Coil Tuning.

Magnitude ____________

1.8 Body Coil Tuning

NOTICE

All endbells MUST be on and fully secured to the magnet to ensure that the RF Body Coil is in the proper position when performing Body Coil Tuning.

Perform Body Coil Tuning per the **Body Coil Tuning** procedure.

Only trained field service personal are permitted to use the RF Field Tuning Kits and are required to pass the training class in MyLearning, GBLSVCSEDTRNG-1549991, “ MR750 Dual Drive RF Field Tuning”.

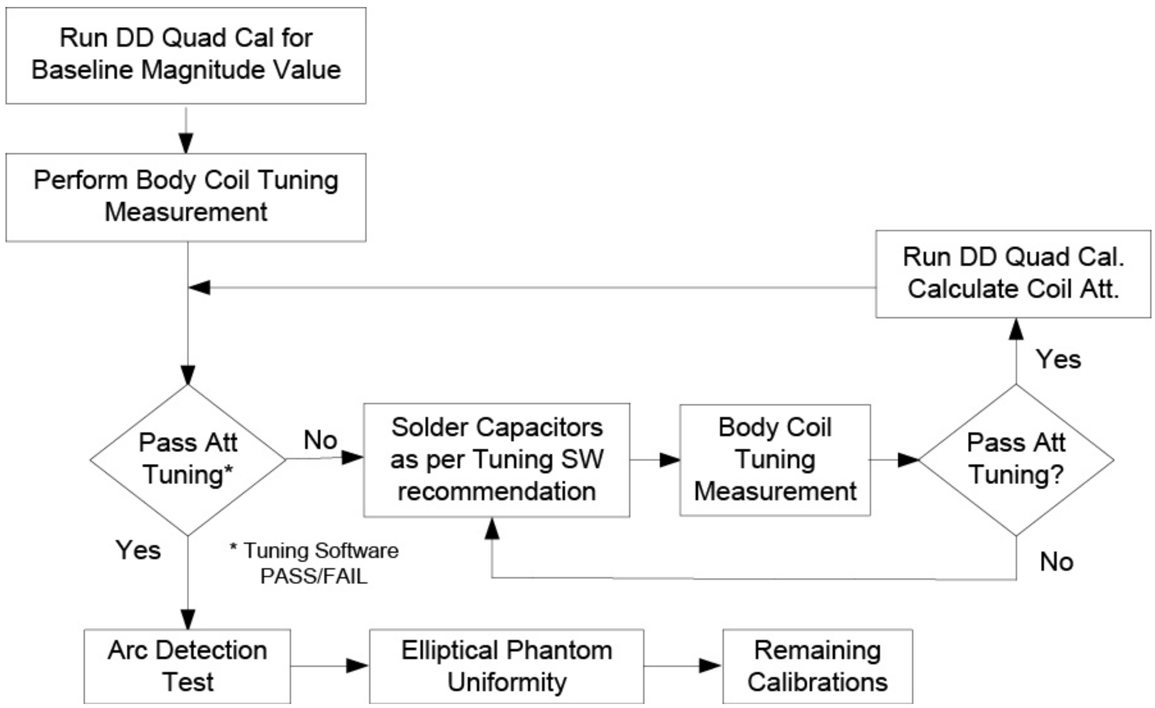
In addition to the RF Field Tuning Kit, the RF Field Tuning software must be installed on a laptop computer prior to beginning the tuning procedure. The software can be ordered by a trained individual. Part number is (5352480-2).

NOTICE

DO NOT FIELD TUNE A BODY COIL UNLESS ABOVE TRAINING IS COMPLETED.

The flow on how to tune the coil is below, however the software that is used to tune the coil should walk you through the process.

Illustration 5-16: Body Coil Tuning Process



1.9 Remaining Calibrations

When body coil tuning is finished, complete the remaining calibrations and Functional checks that appear in the ICW Upgrade mode. They are

- White Pixel
- DQAII Calibration
- LVShim
- Grafidy
- System Gain
- SPT
- EFT

1.10 Perform SavINFO

See service methods Save and Restore Tool

Chapter 6 Appendix

1 Quiet UPM Boards

Table 6-1: MR Quiet Upgrade Master Harvest Test Procedure

MR Quiet Upgrade					
System ID	*	Dispatch Number	*		
Cabinet Serial Number	*	FE Name	*		
		SSO	*		
		Signature	*		
		Test Date	*		
Test Site Name	*	Country	*		
After test completion, send this document with harvested parts for review of test results. Completed test record should be signed and dated. Ensure all cells with * requiring information are completed or marked with "N/A" if not applicable.					
Manual Documentation Number		**Revision Number**	**FE Confirm Rev.**	**If revision number is different, what rev. was used?**	
5670014: Discovery MR450, MR750, Optima MR450w BASE & GEM, MR750w GEM Service Methods		2	*		
Test Number	**Test & Procedure Documentation**	**Expected Results**	**Measured Results**	**Test Result Pass / Fail**	**Specific Failures**
Test					
UPM Diagnostic Procedure Discovery MR450 & MR750, Optima MR450w BASE & GEM, MR750w GEM - *Troubleshooting->Diagnostics->System Function->RF->UPM DIagnostics*					
1	UPM Diagnostics I/O Data	Test completes	Test completes	*Circle One Pass / Fail	
2	UPM Diagnostics Analog Data	Test completes	Test completes	*Circle One Pass / Fail	
Patient Data Removal Verification					
3	Confirm Patient Data has been removed from the hard disc drives	N/A - System does not contain components which hold patient data		N/A	

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